

Dialysis - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses dialysis.

1. Medical Necessity

1. Medically Necessary Procedures

Aetna considers the following procedures medically necessary:

1. Hemodialysis or intermittent peritoneal dialysis for renal failure up to 3 times per week;
2. Hemodialysis and intermittent peritoneal dialysis performed more than 3 times per week for renal failure for hyperkalemia, hypophosphatemia, pregnancy, fluid overload, acute pericarditis, congestive heart failure, pulmonary edema, or severe catabolic state when these conditions are refractory to dialysis 3 times per week;
3. Home hemodialysis when prescribed by a physician for members with end stage renal disease - the following conditions/equipment and supplies are considered medically necessary for administration of hemodialysis in the home:

1. Skilled Nursing

Periodic monitoring of the member's condition by a nurse (skilled nursing visit) in accordance with a care plan that is prepared and periodically reviewed by a physician; *and*

2. Dialysis Equipment and Supplies

1. Equipment: **Note:** Because Medicare primary coverage of the dialysis equipment listed below commences after the person's first 30 months of hemodialysis, this equipment is usually covered by Aetna on a rental basis during the member's first 30 months of hemodialysis, because 30-month's rental is usually less expensive than purchase:
 1. Adjustable reclining chairs, when required as a component of the home dialysis system;
 2. Delivery system accessories:
 1. Blood pumps

2. Heparin infusion pumps
3. Monitoring devices
4. Water purification systems (either a deionization system or a reverse osmosis system are considered medically necessary, but not both in the same member at the same time)
5. Water softening systems for members who have a reverse osmosis purification system if the member's water is of a lesser quality than required for the reverse osmosis purification system;

3. Dialyzers and dialysis delivery systems;

2. Supplies: The following hemodialysis supplies may be considered medically necessary:

1. Activated carbon filters used as a component of water purification systems to remove unsafe concentrations of chlorine or chloramines

2. Antibiotic ointment

3. Blood pressure monitors

4. Blood tubing

5. Centrifuge readacrit (hematocrit measuring equipment)

6. Cleansing agents

7. Dialysates

8. Dialysate testing supplies

9. Fistula cannulation sets

10. Fluid administration sets

11. Gun to secure tubing

12. Heparin

13. Needles

14. Nylon locking ties

15. Reagents (to detect residual traces or cleansing and sterilizing agents)

16. Saline solution components

17. Sterile dressing

18. Sterile saline

19. Sterilizing agents

20. Stethoscope when needed for blood pressure monitoring

21. Syringes

22. Winthrop tubes.

Aetna considers the NxStage System (NxStage Medical, Inc.) or the Tablo Hemodialysis System (Outset Medical, Inc.) portable/mobile hemodialysis machine an equally acceptable alternative to standard hemodialysis machines for medically necessary home hemodialysis.

Aetna considers professional staff to assist home hemodialysis medically necessary for members with end-stage renal disease who meet all of the following criteria:

1. Member is stable on dialysis as shown by meeting the criteria of the National Kidney Foundation; *and*
2. Member has good functioning vascular access; *and*
3. Member has medical contraindications to leaving home for hemodialysis; *and*
4. Member or non-professional care-giver is unable to perform home hemodialysis following hemodialysis training.

2. Peritoneal Dialysis

Aetna considers continuous ambulatory peritoneal dialysis (CAPD) or continuous cycling peritoneal dialysis (CCPD) medically necessary when prescribed by a physician for persons with end-stage renal disease.

3. Other Dialysis Equipment and Supplies

1. Back up equipment supplied in anticipation of the need for substitution or replacement is not considered medically necessary. Rental of equipment is considered medically necessary while member-owned equipment is being repaired.
2. Peridex filter sets are not considered medically necessary for peritoneal dialysis.
3. Short-wave (radiofrequency) diathermy machines are not covered as they are not appropriate for home use. See [CPB 0540 - Heating Devices](#).
4. Spare deionizing tanks are not considered medically necessary since they are essentially a precautionary supply.
5. Ultrafiltration monitors are not considered medically necessary when ultrafiltration is independent of conventional hemodialysis.
Note: Charges for repair and maintenance of rented equipment are included in the rental fees. Charges for repair of rented equipment will be denied as included in the rental charges.
6. Emergency reserve supplies are usually provided when a member initiates dialysis in anticipation of short-term increases in use of supplies or delays in supply delivery. Up to 1 month's supplies in reserve in case of emergency are considered medically necessary; this is a one-time allowance.

4. Hemodialysis Arterio-Venous (AV) Fistulas

1. Aetna considers percutaneous hemodialysis arterio-venous (AV) fistulas (also termed endo-AV fistulas or endoAVFs) (e.g., WavelinQ 4F EndoAVF System (BD Bard, Tempe, AZ); Ellipsys Vascular Access System (Avenu Medical, San Juan Capistrano, CA)) creation medically necessary as an alternative to surgical creation of an AV fistula for persons who require hemodialysis access.
2. Aetna considers angioplasty using FDA-approved plain or drug-coated balloons medically necessary for dysfunctional arterio-venous dialysis fistulae.

2. Experimental, Investigational, or Unproven

1. The Crit-Line In-Line Monitor is a device used to measure a member's hematocrit and oxygen saturation during hemodialysis. This device is considered experimental, investigational, or unproven as there is inadequate evidence of the Crit-Line In-Line Monitor in improving the management of persons receiving hemodialysis. See also [CPB 0373 - Crit-Line In-Line Monitor](#).
2. Aetna considers the following experimental, investigational, or unproven because their effectiveness has not been established:
 1. Bioengineered human acellular vessels for dialysis access;
 2. EchoSure System for predicting arterio-venous fistula (AVF) maturity and impending AVF failure
 3. Interleukin-6 levels for estimation of cardiovascular and all-cause mortality risk in individuals on dialysis;
 4. Peritoneal dialysis as a treatment for heart failure in persons without renal failure;
 5. Prophylactic peritoneal dialysis for improvement of post-operative outcomes in infants/children after cardiac surgery;
 6. The use of multiple-frequency bio-impedance devices for fluid management in persons receiving dialysis;
 7. The use of nasal antibiotic for the prevention of peritonitis in peritoneal dialysis individuals;

8. The use of vitamin E-coated membranes for hemodialysis;
9. VasQ extravascular support device for improvement of surgical fistula outcomes;
10. Wearable hemodialysis machines.

3. Related Policies

1. [CPB 0373 - Crit-Line In-Line Monitor](#)
 2. [CPB 0540 - Heating Devices](#)
 3. [CPB 1028 - Difelikefalin Injection \(Korsuva\)](#)
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CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Hemodialysis:	
CPT codes covered if selection criteria is met:	
36836	Percutaneous arteriovenous fistula creation, upper extremity, single access of both the peripheral artery and peripheral vein, including fistula maturation procedures (eg, transluminal balloon angioplasty, coil embolization) when performed, including all vascular access, imaging guidance and radiologic supervision and interpretation
36837	Percutaneous arteriovenous fistula creation, upper extremity, separate access sites of the peripheral artery and peripheral vein, including fistula maturation procedures (eg, transluminal balloon angioplasty, coil embolization) when performed, including all vascular access, imaging guidance and radiologic supervision and interpretation
90935	Hemodialysis procedure with single evaluation by a physician or other qualified health care professional
90937	Hemodialysis procedure requiring repeated evaluation(s) with or without substantial revision of dialysis prescription
90999	Unlisted dialysis procedure, inpatient or outpatient
99512	Home visit for hemodialysis
CPT codes not covered for indication listed in the CPB:	
<i>Bioengineered human acellular vessels for dialysis access, use of vitamin E-coated membranes for hemodialysis - no specific code</i>	
0876T	Duplex scan of hemodialysis fistula, computer-aided, limited (volume flow, diameter, and depth, including only body of fistula)
83529	Interleukin-6 (IL-6)
HCPCS codes covered if selection criteria is met:	
A4216	Sterile water, saline and/or dextrose, diluent/flush, 10 ml

Code	Code Description
A4217	Sterile water/saline, 500 ml
A4651	Calibrated microcapillary tube, each
A4652	Microcapillary tube sealant
A4657	Syringe, with or without needle, each
A4660	Sphygmomanometer/blood pressure apparatus with cuff and stethoscope
A4663	Blood pressure cuff only
A4670	Automatic blood pressure monitor
A4672	Drainage extension line, sterile, for dialysis, each
A4673	Extension line with easy lock connectors, used with dialysis
A4674	Chemicals/antiseptics solution used to clean/sterilize dialysis equipment, per 8 oz
A4680	Activated carbon filters for hemodialysis, each
A4690	Dialyzer (artificial kidneys), all types, all sizes, for hemodialysis, each
A4706	Bicarbonate concentrate, solution, for hemodialysis, per gallon
A4707	Bicarbonate concentrate, powder, for hemodialysis, per packet
A4708	Acetate concentrate solution, for hemodialysis, per gallon
A4709	Acid concentrate, solution, for hemodialysis, per gallon
A4728	Dialysate solution, non-dextrose, containing, 500 ml
A4730	Fistula cannulation set for hemodialysis, each
A4736	Topical anesthetic for dialysis, per gm
A4737	Injectable anesthetic, for dialysis, per 10 ml
A4740	Shunt accessory, for hemodialysis, any type, each
A4750	Blood tubing, arterial or venous, for hemodialysis, each
A4755	Blood tubing, arterial and venous combined, for hemodialysis, each
A4770	Blood collection tube, vacuum, for dialysis, per 50
A4771	Serum clotting time tube, for dialysis, per 50
A4772	Blood glucose test strips, for dialysis, per 50
A4773	Occult blood test strips, for dialysis, per 50
A4774	Ammonia test strips, for dialysis, per 50
A4802	Protamine sulfate, for hemodialysis, per 50 mg
A4860	Disposable catheter tips for peritoneal dialysis, per 10
A4890	Contracts, repair and maintenance, for hemodialysis equipment
A4911	Drain bag/bottle, for dialysis, each
A4913	Miscellaneous dialysis supplies, not otherwise specified

Code	Code Description
A4918	Venous pressure clamps, for hemodialysis, each
A4927	Gloves, non-sterile, per 100
A4928	Surgical mask, per 20
A4929	Tourniquet for dialysis, each
A4930	Gloves, sterile, per pair
A6010 - A6457	Dressings
C1881	Dialysis access system (implantable)
E1500	Centrifuge, for dialysis
E1510	Kidney, dialysate delivery system kidney machine, pump recirculating, air removal system, flowrate meter, power off, heater and temp control with alarm, IV poles, pressure gauge, concentrate container
E1520	Heparin infusion pump for hemodialysis
E1530	Air bubble detector for hemodialysis, each, replacement
E1540	Pressure alarm for hemodialysis, each, replacement
E1550	Bath conductivity meter for hemodialysis, each
E1560	Blood leak detector for hemodialysis, each, replacement
E1570	Adjustable chair, for ESRD patients
E1575	Transducer protectors/fluid barriers, for hemodialysis, any size, per 10
E1580	Unipuncture control system for hemodialysis
E1590	Hemodialysis machine
E1600	Delivery and/or installation charges for hemodialysis equipment
E1610	Reverse osmosis water purification system, for hemodialysis
E1615	Deionizer water purification system, for hemodialysis
E1620	Blood pump for hemodialysis, replacement
E1625	Water softening system, for hemodialysis
E1629	Tablo hemodialysis system for the billable dialysis service
E1636	Sorbent cartridges, for hemodialysis, per 10
E1699	Dialysis equipment, not otherwise specified
G0299	Direct skilled nursing services of a registered nurse (RN) in the home health or hospice setting, each 15 minutes
G0300	Direct skilled nursing services of a license practical nurse (LPN) in the home health or hospice setting, each 15 minutes
J1643	Injection, heparin sodium (pfizer), not therapeutically equivalent to J1644, per 1000 units
J1644	Injection, heparin sodium, per 1,000 units
J7030	Infusion, normal saline solution, 1,000 cc

Code	Code Description
J7040	Infusion, normal saline solution, sterile (500 ml = 1 unit)
S9123	Nursing care, in the home; by registered nurse, per hour (use for general nursing care only, not to be used when CPT codes 99500-99602 can be used)
S9124	Nursing care, in the home; by licensed practical nurse, per hour
S9335	Home therapy, hemodialysis; administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing services coded separately), per diem
HCPCS codes not covered for indications listed in the CPB:	
C8000	Support device, extravascular, for arteriovenous fistula (implantable)
E1632	Wearable artificial kidney, each
E1635	Compact (portable) travel hemodialyzer system
ICD-10 codes covered if selection criteria are met:	
N18.6	End stage renal disease
Z99.2	Dependence on renal dialysis [not covered for estimation of cardiovascular and all-cause mortality risk]
Peritoneal Dialysis:	
CPT codes covered if selection criteria are met:	
90945	Dialysis procedure other than hemodialysis (eg, peritoneal dialysis, hemofiltration, or other continuous renal replacement therapies), with single physician evaluation [CAPD, CCPD]
90947	Dialysis procedure other than hemodialysis (eg, peritoneal dialysis, hemofiltration, or other continuous renal replacement therapies) requiring repeated physician evaluations, with or without substantial revision of dialysis prescription [CAPD, CCPD]
HCPCS codes covered if selection criteria is met:	
A4216	Sterile water, saline and/or dextrose, diluent/flush, 10 ml
A4217	Sterile water/saline, 500 ml
A4651	Calibrated microcapillary tube, each
A4652	Microcapillary tube sealant
A4653	Peritoneal dialysis catheter anchoring device, belt, each
A4657	Syringe, with or without needle, each
A4660	Sphygmomanometer/blood pressure apparatus with cuff and stethoscope
A4663	Blood pressure cuff only

Code	Code Description
A4670	Automatic blood pressure monitor
A4671	Disposable cyclor set used with cyclor dialysis machine, each
A4672	Drainage extension line, sterile, for dialysis, each
A4673	Extension line with easy lock connectors, used with dialysis
A4674	Chemicals/antiseptics solution used to clean/sterilize dialysis equipment, per 8 oz
A4714	Treated water (deionized, distilled, reverse osmosis) for peritoneal dialysis, per gallon
A4720	Dialysate solution, any concentration of dextrose, fluid volume greater than 249 cc, but less than or equal to 999 cc, for peritoneal dialysis
A4721	Dialysate solution, any concentration of dextrose, fluid volume greater than 999 cc, but less than or equal to 1999 cc, for peritoneal dialysis
A4722	Dialysate solution, any concentration of dextrose, fluid volume greater than 1999 cc, but less than or equal to 2999 cc, for peritoneal dialysis
A4723	Dialysate solution, any concentration of dextrose, fluid volume greater than 2999 cc, but less than or equal to 3999 cc, for peritoneal dialysis
A4724	Dialysate solution, any concentration of dextrose, fluid volume greater than 3999 cc, but less than or equal to 4999 cc, for peritoneal dialysis
A4725	Dialysate solution, any concentration of dextrose, fluid volume greater than 4999 cc, but less than or equal to 5999 cc, for peritoneal dialysis
A4726	Dialysate solution, any concentration of dextrose, fluid volume greater than 5999 cc
A4728	Dialysate solution, non-dextrose, containing, 500 ml
A4736	Topical anesthetic for dialysis, per gm
A4737	Injectable anesthetic for dialysis, per 10 ml
A4760	Dialysate solution test kit, for peritoneal dialysis, any type, each
A4765	Dialysate concentrate, powder, additive for peritoneal dialysis, per packet
A4766	Dialysate concentrate, solution, additive for peritoneal dialysis, per 10 ml
A4770	Blood collection tube, vacuum, for dialysis, per 50
A4771	Serum clotting time tube, for dialysis, per 50
A4772	Blood glucose test strips, for dialysis, per 50
A4773	Occult blood test strips, for dialysis, per 50
A4774	Ammonia test strips, for dialysis, per 50
A4860	Disposable catheter tips for peritoneal dialysis, per 10
A4911	Drain bag/bottle, for dialysis, each
A4913	Miscellaneous dialysis supplies, not otherwise specified
A4918	Venous pressure clamps, for hemodialysis, each
A4927	Gloves, non-sterile, per 100

Code	Code Description
A4928	Surgical mask, per 20
A4929	Tourniquet for dialysis, each
A4930	Gloves, sterile, per pair
A6010 - A6457	Dressings
C1881	Dialysis access system (implantable)
E1500	Centrifuge, for dialysis
E1510	Kidney, dialysate delivery system kidney machine, pump recirculating, air removal system, flowrate meter, power off, heater and temp control with alarm, IV poles, pressure gauge, concentrate container
E1570	Adjustable chair, for ESRD patients
E1592	Automatic intermittent peritoneal dialysis system
E1594	Cycler dialysis machine for peritoneal dialysis
E1630	Reciprocating peritoneal dialysis system
E1634	Peritoneal dialysis clamps, each
E1699	Dialysis equipment, not otherwise specified
G0299	Direct skilled nursing services of a registered nurse (RN) in the home health or hospice setting, each 15 minutes
G0300	Direct skilled nursing services of a license practical nurse (LPN) in the home health or hospice setting, each 15 minutes
S9123	Nursing care, in the home; by registered nurse, per hour
S9124	Nursing care, in the home; by licensed practical nurse, per hour
S9335	Home therapy, hemodialysis; administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing services coded separately), per diem
S9339	Home therapy; peritoneal dialysis, administrative services, professional pharmacy services, care coordination and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem
HCPCS codes not covered for indications listed in the CPB:	
<i>Multiple-frequency bio-impedance devices - no specific code:</i>	
A4870	Plumbing and/or electrical work for home hemodialysis equipment
E1637	Hemostats, each
E1639	Scale, each
Other HCPCS codes related to the CPB:	

Code	Code Description
C7513	Dialysis circuit, introduction of needle(s) and/or catheter(s), with diagnostic angiography of the dialysis circuit, including all direct puncture(s) and catheter placement(s), injection(s) of contrast, all necessary imaging from the arterial anastomosis and adjacent artery through entire venous outflow including the inferior or superior vena cava, fluoroscopic guidance, with transluminal balloon angioplasty of central dialysis segment, performed through dialysis circuit, including all required imaging, radiological supervision and interpretation, image documentation and report
C7514	Dialysis circuit, introduction of needle(s) and/or catheter(s), with diagnostic angiography of the dialysis circuit, including all direct puncture(s) and catheter placement(s), injection(s) of contrast, all necessary imaging from the arterial anastomosis and adjacent artery through entire venous outflow including the inferior or superior vena cava, fluoroscopic guidance, with all angioplasty in the central dialysis segment, and transcatheter placement of intravascular stent(s), central dialysis segment, performed through dialysis circuit, including all required imaging, radiological supervision and interpretation, image documentation and report
C7515	Dialysis circuit, introduction of needle(s) and/or catheter(s), with diagnostic angiography of the dialysis circuit, including all direct puncture(s) and catheter placement(s), injection(s) of contrast, all necessary imaging from the arterial anastomosis and adjacent artery through entire venous outflow including the inferior or superior vena cava, fluoroscopic guidance, with dialysis circuit permanent endovascular embolization or occlusion of main circuit or any accessory veins, including all required imaging, radiological supervision and interpretation, image documentation and report
C7530	Dialysis circuit, introduction of needle(s) and/or catheter(s), with diagnostic angiography of the dialysis circuit, including all direct puncture(s) and catheter placement(s), injection(s) of contrast, all necessary imaging from the arterial anastomosis and adjacent artery through entire venous outflow including the inferior or superior vena cava, fluoroscopic guidance, with transluminal balloon angioplasty, peripheral dialysis segment, including all imaging and radiological supervision and interpretation necessary to perform the angioplasty and all angioplasty in the central dialysis segment, with transcatheter placement of intravascular stent(s), central dialysis segment, performed through dialysis circuit, including all imaging, radiological supervision and interpretation, documentation and report
E1632	Wearable artificial kidney, each
G0491	Dialysis procedure at a medicare certified esrd facility for acute kidney injury without esrd
G0492	Dialysis procedure with single evaluation by a physician or other qualified health care professional for acute kidney injury without esrd
J1642	Injection, heparin sodium, (Heparin Lock Flush), per 10 units
ICD-10 codes covered if selection criteria are met:	
E83.39	Other disorders of phosphorous metabolism [Hypophosphatemia]
E87.5	Hyperkalemia
E87.70 - E87.79	Fluid overload

Code	Code Description
I30.0 - I30.9	Acute pericarditis
I50.1 - I50.9	Heart failure
J81.0 - J81.1	Pulmonary edema
N18.6	End stage renal disease
000.0 - 09A.53	Pregnancy, childbirth and the puerperium
R54	Age-related physical debility [severe catabolic state]
Z33.1	Pregnant state, incidental
Z33.3	Pregnant state, gestational carrier
Z3A.00 - Z3A.49	Weeks of gestation
<i>Drug-coated balloon angioplasty:</i>	
CPT codes covered if selection criteria are met:	
<i>Drug-coated balloon angioplasty - no specific code</i>	
ICD-10 codes covered if selection criteria are met:	
T82.590A - T82.590S	Other mechanical complication of surgically created arteriovenous fistula [Dysfunctional]
T82.858A - T82.858S	Stenosis of other vascular prosthetic devices, implants and grafts [dialysis access stenosis]

Background

This policy on dialysis was adapted from Medicare DMERC Local Medical Policy.

Hemodialysis or intermittent peritoneal dialysis (PD) is necessary up to 3 times per week. Hemodialysis and intermittent PD performed more than 3 times per week may be necessary for hyperkalemia, hypophosphatemia, pregnancy, fluid overload, acute pericarditis, congestive heart failure, pulmonary edema, or severe catabolic state when these conditions are refractory to dialysis 3 times per week.

Home Hemodialysis

NxStage System

NxStage Medical, Inc., a subsidiary of Fresenius Medical Care, offers the NxStage System One portable hemodialysis machine, which is FDA cleared for the treatment of acute and chronic renal failure or fluid overload using hemofiltration, hemodialysis, and/or ultrafiltration, in an acute or chronic care facility. The NxStage System is also indicated for home hemodialysis, including home nocturnal hemodialysis. All treatments

must be administered under physician's prescription, and must be performed by a trained and qualified person, considered to be competent in the use of this device by the prescribing physician (FDA, 2014).

More frequent hemodialysis has the potential to improve survival as well as quality of life of patients with chronic kidney disease. New means of delivering hemodialysis are being explored. Kraus et al (2007) carried out a feasibility study to examine the safety of center-based versus home-based daily hemodialysis with the NxStage System One portable hemodialysis device. These investigators also performed a retrospective analysis to determine if clinical effects previously associated with short-daily dialysis were also seen using this novel device. They conducted a prospective, 2-treatment, 2-period, open-label, cross-over study of in-center hemodialysis versus home hemodialysis in 32 patients treated at 6 centers. The 8-week In-Center Phase (6 days/week) was followed by a 2-week transition period and then followed by the 8-week Home Phase (6 days/week). These researchers retrospectively collected data on hemodialysis treatment parameters immediately preceding the study in a subset of patients. Twenty-six out of 32 patients (81 %) successfully completed the study. Successful delivery of at least 90 % of prescribed fluid volume (primary endpoint) was achieved in 98.5 % of treatments in-center and 97.3 % at home. Total effluent volume as a percentage of prescribed volume was between 94 % and 100 % for all study weeks. The composite rate of intra-dialytic and inter-dialytic adverse events per 100 treatments was significantly higher for the In-Center Phase (5.30) compared with the Home Phase (2.10; $p = 0.007$). Compared with the period immediately preceding the study, there were reductions in blood pressure, anti-hypertensive medications, and inter-dialytic weight gain. The authors concluded that daily home hemodialysis with a small, easy-to-use hemodialysis device is a viable dialysis option for end-stage renal disease (ESRD) patients capable of self/partner-administered dialysis.

The Canadian Agency for Drugs and Technologies in Health's report on portable home hemodialysis for kidney failure (Scott, 2007) stated that while home hemodialysis is less costly than conventional in-center programs, it is unknown if these savings extend to portable devices. Presently, the NxStage System One is the only really portable hemodialysis system licensed in the United States. It weighs 30 kg and is the size of an older style computer monitor. This device operates on standard electric current; does not require any water supply, plumbing, or disinfection; and is portable enough for travel. The NxStage System consists of a computer-controlled delivery unit and a disposable cartridge containing the dialyser and fluid circuits. The dialysate comes in sterile, pre-mixed bags, which eliminates the need for a water purification system. An optional accessory can produce dialysate from purified home tap water. Another manufacturer, Home Dialysis Plus Ltd., has developed a portable hemodialysis machine that is smaller and more efficient than existing systems. The Home Dialysis Plus machine weighs approximately 14 kg and is the size of a large suitcase.

The CADTH (Scott, 2007) stated that the only publications on the NxStage System are brief reports and conference presentations of case series studies, some of which pooled results from different dialysis machines. One anecdotal report and three conference abstracts provided separate results. Furthermore, it is not yet clear if the use of the portable hemodialysis machine (e.g. the NxStage System) improves long-term survival and quality of life.

The CADTH report noted that the FREEDOM (Following Rehabilitation, Economics and Everyday-Dialysis Outcome Measurements) Study may address this lack of evidence by comparing clinical outcomes and cost-effectiveness data from 500 patients on NxStage daily hemodialysis with a matched conventional in-centre hemodialysis cohort from the U.S. Renal Data System database (Scott, 2007). The results of this and other ongoing trials will influence the uptake of portable hemodialysis devices.

An assessment by the National Horizon Scanning Centre (NHSC, 2007) found a lack of evidence for the NxStage and other portable systems that utilize new means of generating dialysate from tap water. An assessment of the NxStage System by the Adelaide Health Technology Assessment National Horizon Scanning Unit (Purins and Hiller, 2008) found that the evidence for the NxStage System was from small, low to medium quality studies.

More recently, and assessment from the National Health Service (Gossage-Worrall, et al., 2010) found that "the evidence for the use of portable hemodialysis devices is limited. The available evidence consists of comparison studies, case series, poster presentations, product specific reviews and editorials.... There is a paucity of evidence on the use of this device in relation to its portability."

A report by the National Horizon Scanning Centre (NHSC, 2012) found no published evidence comparing the compact transportable home hemodialysis machines (NxStage or Selfcare+) to standard hemodialysis machines. "Research comparing the costs associated with transportable machines and standard home haemodialysis machines would be needed to further understand and quantify any potential benefits or risks of these machines. The results of the Freedom study and further research into adolescents and children are awaited, as are trials of the Selfcare+ system. Further long-term studies of various possible treatment regimes on these transportable machines compared with standard machines are needed to assess whether they offer improvements in health outcomes for patients."

Jaber et al (2009) noted that conventional thrice-weekly hemodialysis (HD) has limited the ability to generate further improvements in patient quality of life, morbidity, and mortality. Daily HD (DHD) offers the promise of providing clinical and economical benefits. The authors reported that the objectives of the FREEDOM Study, and observational study with a lack of control arm, were to evaluate outcomes of DHD (6 times/week) with the NxStage System One device. The DHD group will include up to 500 participants at 70 clinical sites, enrolling for 3 years with a minimum of 1-year follow-up. Study candidates include adult patients (age greater than or equal to 18 years) with ESRD who are considered suitable candidates for DHD with the NxStage System One device by the treating physician and who have Medicare as their primary insurance payer. The control group will consist of a matched thrice-weekly in-center HD cohort derived from the U.S. Renal Data System database using a 10:1 ratio, totaling 5,000 patients. The primary intent-to-treat analysis compares hospitalization days/patient-year between the DHD and thrice-weekly HD groups. Other outcomes recorded in both groups include non-treatment-related medical expenditures. In addition, in the DHD cohort, changes in quality-of-life measures (baseline, 4 and 12 months, and every 6 months thereafter); urea kinetics; parameters related to anemia, bone and mineral metabolism, and nutrition; vascular access interventions; and use of medications will be examined. This authors concluded that this study has the potential to elucidate the health and economic benefits of DHD and complement results of current clinical trials.

In an interim report from the FREEDOM study, Jaber et al (2010) reported on the impact of daily hemodialysis on depressive symptoms and post-dialysis recovery time. The authors found that daily hemodialysis is associated with long-term improvement in depressive symptoms and postdialysis recovery time. In this interim report, as part of an a priori planned analysis, the investigators examined the long-term impact of daily hemodialysis on depressive symptoms, measured using the Beck Depression Inventory (BDI) survey, and post-dialysis recovery time, measured using a previously validated questionnaire, in adult patients initiating daily hemodialysis. The BDI survey and postdialysis recovery time question were administered at baseline, and changes were assessed at months 4 and 12. The investigators reported that 239 participants were enrolled (intention-to-treat cohort) and 128 completed the study (per-protocol cohort). Mean age was 52 years, 64 % were men, 55 % had an arterio-venous (AV) fistula, and 90 % transitioned from in-center hemodialysis therapy. In the per-protocol cohort, there was a significant decrease in mean BDI score over 12 months (11.2 [95 % confidence interval [CI]: 9.6 to 12.9] versus 7.8 [95 % CI: 6.5 to 9.1]; $p < 0.001$). For robustness, the intention-to-treat analysis was performed, yielding similar results. The percentage of patients with depressive symptoms (BDI score greater than 10) significantly decreased during 12 months (41 % versus 27 %; $p = 0.03$). Similarly, in the per-protocol cohort, there was a significant decrease in post-dialysis recovery time over 12 months (476 [95 % CI: 359 to 594] versus 63 minutes [95 % CI: 32 to 95]; $p < 0.001$). The intention-to-treat analysis yielded similar results. The percentage of patients experiencing prolonged post-dialysis recovery time (greater than or equal to 60 minutes) also significantly decreased (81 % versus 35 %; $p = 0.001$).

A retrospective study using a matched population-based cohort by Weinhandl et al (2012) suggests that relative to thrice-weekly in-center hemodialysis, daily home hemodialysis with the NxStage System associates with modest improvements in survival. The investigators used a matched-cohort design to assess relative mortality in daily home hemodialysis using the NxStage System and thrice-weekly in-center hemodialysis patients between 2005 and 2008. The investigators matched 1,873 home hemodialysis patients with 9,365 in-center patients (i.e., 1:5 ratio) selected from the prevalent population in the U.S. Renal Data System database. Matching variables included first date of follow-up, demographic characteristics, and measures of disease severity. The cumulative incidence of death was 19.2 % and 21.7 % in the home hemodialysis and in-center patients, respectively. In the intention-to-treat analysis, home hemodialysis using the NxStage

was associated with a 13 % lower risk for all-cause mortality than in-center hemodialysis (hazard ratio [HR], 0.87; 95 % CI: 0.78 to 0.97). Cause-specific mortality HRs were 0.92 (95 % CI: 0.78 to 1.09) for cardiovascular disease, 1.13 (95 % CI: 0.84 to 1.53) for infection, 0.63 (95 % CI: 0.41 to 0.95) for cachexia/dialysis withdrawal, 1.06 (95 % CI: 0.81 to 1.37) for other specified cause, and 0.59 (95 % CI: 0.44 to 0.79) for unknown cause. Findings were similar using as-treated analyses. The investigators reported that they did not detect statistically significant evidence of heterogeneity of treatment effects in subgroup analyses.

Walker et al (2014) sought comparative cost-effectiveness studies of home versus facility HD for people with end-stage kidney failure (ESKF). These investigators conducted a systematic review of literature from January 2000 to March 2014. Studies were included if they provided comparative information on the costs, health outcomes and cost-effectiveness ratios of home HD and facility HD. They searched medical and health economic databases using MeSH headings and text words for economic evaluation and hemodialysis. A total of 6 studies of economic evaluations that compared home to facility HD were identified – 2 studies compared home nocturnal HD, 1 home nocturnal and daily home HD, and 3 compared contemporary home HD to facility HD. Overall, these studies suggested that contemporary home HD modalities are less costly and more effective than facility HD. Home HD start-up costs tend to be higher in the short-term, but these are offset by cost savings over the longer term. The authors concluded that contemporaneous dialysis modalities including nocturnal and daily home HD are cost-effective or cost-saving compared to facility-based HD. This result is largely driven by lower staff costs, and better health outcomes for survival and quality of life.

Expanding the proportion of HD patients managed at home is likely to produce cost savings.

Palmer et al (2014) stated that home hemodialysis is associated with improved survival and quality of life (QOL) in uncontrolled studies. However, relative benefits and harms of home versus in-center hemodialysis in randomized controlled trials (RCTs) are uncertain. In a Cochrane review, these investigators evaluated the benefits and harms of home hemodialysis versus in-center hemodialysis in adults with end-stage kidney disease (ESKD). The Cochrane Renal Group's Specialised Register was searched up to October 31, 2014; RCTs of home versus in-center hemodialysis in adults with ESKD were included. Data were extracted by 2 investigators independently. Study risk of bias and other patient-centered outcomes were extracted. Insufficient data were available to conduct meta-analyses. These researchers identified a single cross-over RCT (enrolling 9 participants) that compared home hemodialysis (long hours: 6 to 8 hours, 3 times/week) with in-center hemodialysis (short hours: 3.5 to 4.5 hours, 3 times/weeks) for 8 weeks in prevalent home hemodialysis patients. Outcome data were limited and not available for the end of the first phase of treatment in this cross-over study which was at risk of bias due to differences in dialysate composition between the 2 treatment comparisons. Overall, home hemodialysis reduced 24 hour ambulatory blood pressure and improved uremic symptoms, but increased treatment-related burden of disease and interference in social activities. Insufficient data were available for mortality, hospitalization or dialysis vascular access complications or treatment durability. The authors concluded that insufficient randomized data were available to determine the effects of home hemodialysis on survival, hospitalization, and QOL compared with in-center hemodialysis. They stated that given the consistently observed benefits of home hemodialysis on QOL and survival in uncontrolled studies, and the low prevalence of home hemodialysis globally, RCTs evaluating home hemodialysis would help inform clinical practice and policy.

Tablo Hemodialysis System

Outset Medical, Inc. offers the Tablo Hemodialysis System, which is FDA cleared for use in patients with acute and/or chronic renal failure, with or without ultrafiltration, in an acute or chronic care facility. Treatments must be administered under physician's prescription, with a trained individual available as needed who is considered competent in the use of the device by the prescribing physician. The Tablo Hemodialysis System is also indicated for use in the home (FDA, 2020). The Tablo Hemodialysis System is a compact mobile unit on wheels that includes wireless connectivity and touchscreen guidance.

In an investigational device exemption (IDE) study, Plumb et al (2020) evaluated the safety and efficacy of the Tablo hemodialysis system managed in-center by health care professionals and in-home by patients and/or caregivers. The authors conducted a prospective, multicenter, open-label, crossover trial comparing in-center and in-home

hemodialysis using Tablo. There were 4 treatment periods during which hemodialysis was prescribed 4 times per week: 1-week Run-In, 8-week In-Center, 4-week Transition, and 8-week In-Home. The primary efficacy endpoint was weekly standard Kt/Vurea ≥ 2.1 . The secondary efficacy endpoint was delivery of ultrafiltration (UF) within 10% of prescribed UF. Thirty participants enrolled and 28 completed all trial periods. The authors found that adherence to the protocol requirement of 4 treatments per week was 96% in-center and 99% in-home. The average prescribed and delivered session lengths were 3.4 hours for both the In-Center and the In-Home periods. The primary efficacy endpoint for the intention-to-treat cohort was achieved in 199/200 (99.5%) of measurements during the In-Center period and 168/171 (98.3%) In-Home. The average weekly standard Kt/Vurea was 2.8 in both periods. The secondary efficacy UF endpoint was achieved in the ITT cohort in 94% in both in-center and in-home. Two prespecified adverse events (AEs) occurred during the In-Center period and 6 in the In-Home period. None of the AEs were deemed by investigators as related to Tablo. The median resolution time of alarms was 8 seconds in-center and 5 seconds in-home. The authors concluded that the primary and secondary efficacy and safety endpoints were achieved during both In-Center and In-Home trial periods. The authors state that their study confirms that Tablo is safe and effective for home hemodialysis use. The authors note several limitations to their study, which included that the number of patients was relatively small, and the duration of each study period was relatively brief. The average age of the patients was younger than the average age of the patients receiving dialysis in the United States, although consistent with other studies on in-home hemodialysis. Given that the willingness to do hemodialysis at home was an inclusion criterion, study participants were more likely to have an ease with medical technology and an interest in self-care which may also contribute to selection bias. Other, less motivated patients may not have achieved similar results.

Bioengineered Human Acellular Vessels for Dialysis Access

Lawson and colleagues (2016) stated that for patients with end-stage renal disease (ESRD) who are not candidates for fistula, dialysis access grafts are the best option for chronic HD. However, polytetrafluoroethylene arterio-venous grafts are prone to thrombosis, infection, and intimal hyperplasia at the venous anastomosis. These researchers developed and tested a bioengineered human acellular vessel as a potential solution to these limitations in dialysis access. They performed 2 single-arm phase II clinical trials at 6 centers in the US and Poland. These investigators enrolled adults with ESRD. A novel bioengineered human acellular vessel was implanted into the arms of patients for HD access. Primary end-points were safety (freedom from immune response or infection, aneurysm, or mechanical failure, and incidence of adverse events [AEs]), and efficacy as assessed by primary, primary assisted, and secondary patencies at 6 months.

All patients were followed-up for at least 1 year, or had a censoring event. Human acellular vessels were implanted into 60 patients; mean follow-up was 16 months (SD 7.6).

One vessel became infected during 82 patient-years of follow-up. The vessels had no dilatation and rarely had post-cannulation bleeding. At 6 months, 63 % (95 % CI: 47 to 72) of patients had primary patency, 73 % (57 to 81) had primary assisted patency, and 97 % (85 to 98) had secondary patency, with most loss of primary patency because of thrombosis. At 12 months, 28 % (17 to 40) had primary patency, 38 % (26 to 51) had primary assisted patency, and 89 % (74 to 93) had secondary patency. The authors concluded that bioengineered human acellular vessels appeared to provide safe and functional HD access, and warrant further study in RCTs.

Combined Blockade of Renin-Angiotensin-Aldosterone System

Li and co-workers (2018) stated that full blockade of the renin-angiotensin-aldosterone system (RAAS) is believed to decrease morbidity and mortality of patients with chronic kidney disease (CKD). In non-dialysis patients, combined RAAS blockade with 2 different RAAS blockers causes more AEs without improving survival, but its role in maintenance dialysis patients is still unclear. These investigators conducted a systematic review and mediation analysis to examine the safety and efficacy of combined RAAS blockade in dialysis patients. Comprehensive search was conducted in PubMed, Embase, Web of Science and Cochrane Library database to June 2017 to identify relevant studies. Studies comparing combined with single RAAS blockade and reporting all-cause death, cardiovascular death, hypotension or hyperkalemia in dialysis patients were included.

Effect sizes were calculated with randomized effects model and summarized as odd ratios (OR). A total of 9 studies with 13,050 dialysis patients were included. Compared with single blockade, combined blockade significantly reduced all-cause mortality (OR 0.71, 95 % CI: 0.54 to 0.93, $p = 0.01$), while cardiovascular mortality remained unchanged (OR

0.85, 95 % CI: 0.45 to 1.59, $p = 0.61$). Combined blockade tended to increase odd of hypotension but not odd of hyperkalemia (OR 1.54, 95 % CI: 1.00 to 2.38, $p = 0.05$; OR 0.89, 95 % CI: 0.76 to 1.05, $p = 0.17$). Further mediation analysis indicated that hypotension might exert a suppression effect on the survival benefit of angiotensin-converting enzyme (ACE) inhibitor plus angiotensin receptor blocker (ARB) treatment on cardiovascular mortality. The authors concluded that combined RAAS blockade might be a promising treatment in dialysis patients to further reduce mortality if blood pressure (BP) was well-controlled.

Drug-Coated Balloon Angioplasty for Dialysis Access Stenosis

Wee and colleagues (2019) noted that arterio-venous fistulas for patients undergoing HD are at high risk of stenosis. Despite conventional balloon angioplasty (CBA), re-stenosis rates are high. The use of a drug-coated balloon (DCB) may offer an alternative to reduce re-stenosis. This study was carried out in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. An electronic search on Medline, Embase, and the Cochrane Library was conducted to identify articles evaluating DCB angioplasty for patients with HD access stenosis; RRs of primary patency were pooled, and relevant subgroup and sensitivity analyses were conducted.

There were 17 studies (8 RCTs, 9 cohort studies) included, comprising a total of 1,113 stenotic dialysis accesses, of which 54.7 % underwent DCB angioplasty and 45.3 % underwent CBA. There was a significantly superior 6-month (RR, 0.57; 95 % CI: 0.44 to 0.74; $p < 0.00001$; $I^2 = 62$ %) and 12-month (RR, 0.73; 95 % CI: 0.63 to 0.84; $p < 0.0001$; $I^2 = 53$ %) primary patency in the DCB angioplasty group in comparison to the CBA group (71.0 % versus 49.2 % at 6 months; 44.2 % versus 20.6 % at 12 months). Subgroup analyses of study design (RCTs, cohort studies) showed similar trends. Sensitivity analyses by excluding 1 poor-quality RCT and those employing the cross-over analysis design also showed similar results. Studies investigating central venous stenosis showed significantly better 6-month (RR, 0.57; 95 % CI: 0.41 to 0.79; $p = 0.0009$; $I^2 = 67$ %) and 12-month (RR, 0.69; 95 % CI: 0.56 to 0.85; $p = 0.0004$; $I^2 = 64$ %) primary patency in the DCB angioplasty group in comparison to the CBA group. The pooled rate of minor complications was low in both the DCB (1.1 %) and CBA (0.9 %) groups. The authors concluded that DCB angioplasty appeared to be a better and safe alternative to CBA in treating patients with HD stenosis in terms of 6- and 12-month primary patency.

Moreover, these researchers stated that a larger trial is needed to establish these findings.

Liao and associates (2020) noted that re-stenosis remains a significant problem in endovascular therapy for HD vascular access; and DCB angioplasty decreases re-stenosis in peripheral and coronary artery diseases. In a systematic review and meta-analysis, these researchers examined the patency outcomes following DCB angioplasty, as compared to CBA for the stenosis of HD vascular access. They carried out a comprehensive search in the Medline, Embase, and CENTRAL databases to identify eligible RCTs evaluating DCB angioplasty for HD vascular access dysfunction. The primary end-point was the 6-month target lesion primary patency and the secondary end-points were 12-month target lesion primary patency and procedure-related complications; RR were pooled and relevant subgroups were analyzed separately. A total of 11 RCTs comprised of 487 patients treated with DCB angioplasty and 489 patients treated with CBA were included. There were no significant differences in the target lesion primary patency at 6 months [RR, 0.75; 95 % CI: 0.56 to 1.01; $p = 0.06$] and at 12 months (RR 0.89; 95 % CI: 0.79 to 1.00; $p = 0.06$).

The absence of benefit for the DCB group remained, even in the arterio-venous fistula subgroup or the subgroup of studies excluding central vein stenosis. The risk of procedure-related complication did not differ between the 2 groups (RR 1.00; 95 % CI: 0.98 to 1.02; $p = 0.95$). The authors concluded that DCB angioplasty did not demonstrate significant patency benefit for the treatment of HD vascular access dysfunction; wide variations in patency outcomes across studies were noted. Moreover, these researchers stated that further studies focusing on specific types of access or lesions are needed to clarify the value of DCB for HD vascular access.

An UpToDate review on techniques for angioplasty of the arteriovenous hemodialysis access (Beathard, 2020) states that small trials and single-center observational studies had suggested a benefit for drug-coated balloon angioplasty compared with standard balloon angioplasty. The review states, however, that in multicenter trials, drug-eluting balloon angioplasty has not improved patency rates (citing Abdul Salim, et al., 2020; Moreno-Sánchez, et al., 2020).

Han, et al. (2021) performed a systematic review and meta-analysis of randomized

controlled trials (RCTs) was to determine the effectiveness and safety of drug coated balloon (DCB) angioplasty compared with uncoated plain balloon (PB) angioplasty in treating arteriovenous access stenosis. MEDLINE, Embase, Scopus, and the Cochrane Central Register of Controlled Trials were searched for RCTs comparing paclitaxel coated DCB and PB angioplasty for arteriovenous access stenosis. The last date of the literature search was 31 December 2020. Risk of bias of the retrieved studies was assessed with the Cochrane Collaboration tool for assessing risk of bias (RoB 2.0). The random effects model was used to estimate the risk of loss of target lesion patency (six and 12 months) and circuit patency (six and 12 months). Procedure related adverse events and mortality rate were also compared. Patency results were pooled using the time to event meta-analytical method and the quality of evidence was assessed according to the GRADE approach. Sixteen eligible trials, including 1 682 lesions, were included in the quantitative analysis for the efficacy and safety of paclitaxel coated DCBs. DCBs were associated with a lower risk of loss of target lesion patency at six months (HR 0.53, 95% CI 0.42 - 0.66) and 12 months (HR 0.60, 95% CI 0.47 - 0.76), and were also associated with improved six and 12 month circuit patency. Overall quality of evidence was moderate to low. Procedural complications were rare, and the risk of death up to 12 months was similar between the two groups (OR 1.03, 95% CI 0.68 - 1.56). The authors concluded that paclitaxel coated DCBs reduced the risk of loss of target lesion patency and circuit patency in arteriovenous access stenosis compared with PBs. The authors stated that, considering the heterogeneity of the included trials, there is a need to investigate optimal treatment regimens regarding drug dose and agent of the DCB and the treatment procedure.

Measurement of Serum Uric Acid for Prediction of Cardiovascular or All-Cause Mortality in Hemodialysis Patients

Rohn and associates (2020) stated that in the general population, hyperuricemia is associated with increased morbidity and mortality. Data on this association in hemodialysis patients is controversial. Moreover, it remains elusive whether serum uric acid (SUA) lowering therapy is associated with mortality. These researchers carried out a retrospective analysis of 601 patients on chronic hemodialysis therapy in 5 outpatient centers with a maximum follow-up of 100 months and a mean follow-up of 41 months. Death was defined as primary endpoint. Cumulative survival was analyzed by Kaplan-Meier analysis and Cox regressions adjusted for age. Cumulative survival rates were higher for those subjects with a higher than median SUA concentration both based on mean annual and baseline measurements ($p < 0.05$ each). There was no survival difference anymore after adjustment for age ($p > 0.05$ each). Stratification for SUA lowering therapy (allopurinol/febuxostat) had no impact on cumulative survival, neither in Kaplan Meier nor in Cox regression analyses ($p > 0.05$ each). Furthermore, Cox regression analysis excluded an increased cardiovascular mortality in subjects with hyperuricemia. The authors concluded that in contrast to the general population, hyperuricemia was not associated with increased mortality in patients undergoing hemodialysis. Moreover, xanthine oxidase inhibition was not associated with a survival benefit in this analysis. These data did not support the use of SUA lowering medication in hemodialysis patients with asymptomatic hyperuricemia.

Wang and colleagues (2021) noted that studies have reported inconsistent findings regarding the association between SUA levels and mortality in hemodialysis patients. In a meta-analysis, these researchers examined if higher SUA values comprised a risk factor of cardiovascular or all-cause mortality in maintenance hemodialysis patients. PubMed, Embase and the Cochrane library were searched up to August 31, 2020 for the longitudinal studies that examined the association between the elevated SUA and cardiovascular or all-cause mortality risk in maintenance hemodialysis patients. Pooled adjusted HR and corresponding 95 % CI were calculated using a random-effects model. These investigators included 10 studies with an overall sample of 264,571 patients with hemodialysis in this meta-analysis. Patients with the highest SUA were associated with a decreased risk of cardiovascular mortality (hazard ratio [HR] = 0.72, 95 % CI: 0.59 to 0.87) compared with patients with the lowest SUA after adjustment for potential confounders in a random effects model. Moreover, for each increase of 1 mg/dL of SUA, the overall risks of all-cause and cardiovascular mortality decreased by 6 % and 9 %, respectively (HR = 0.94, 95% CI: 0.90 to 0.99; HR = 0.91, 95 % CI: 0.89 to 0.94). The authors concluded that elevated SUA levels were strongly and independently associated with lower risk of cardiovascular mortality in maintenance hemodialysis patients. Moreover, these researchers stated that further investigations, especially RCTs, are needed to examine if high SUA levels is an independent risk factor of all-cause mortality in hemodialysis

patients.

Multiple-Frequency Bio-Impedance Devices for Fluid Management in Persons Receiving Dialysis

Scotland and associates (2018) stated that CKD is a long-term condition requiring treatment such as conservative management, kidney transplantation or dialysis. To optimize the volume of fluid removed during dialysis (to avoid under-hydration or over-hydration), individuals are assigned a "target weight", which is commonly assessed using clinical methods, such as weight gain between dialysis sessions, pre- and post-dialysis BP and patient-reported symptoms. However, these methods are not precise, and measurement devices based on bio-impedance technology are increasingly used in dialysis centers. Current evidence on the role of bio-impedance devices for fluid management in people with CKD receiving dialysis is limited. In a systematic review, these investigators evaluated the clinical effectiveness and cost-effectiveness of multiple-frequency bio-impedance devices versus standard clinical assessment for fluid management in people with CKD receiving dialysis. These researchers searched major electronic databases [e.g., Medline, Medline In-Process & Other Non-Indexed Citations, Embase, Science Citation Index and Cochrane Central Register of Controlled Trials (CENTRAL)] conference abstracts and ongoing studies. There were no date restrictions. Searches were undertaken between June and October 2016. Evidence was considered from RCTs comparing fluid management by multiple-frequency bio-impedance devices and standard clinical assessment in people receiving dialysis, and non-randomized studies evaluating the use of the devices for fluid management in people receiving dialysis. One reviewer extracted data and assessed the risk of bias of included studies. A second reviewer cross-checked the extracted data. Standard meta-analyses techniques were used to combine results from included studies. A Markov model was developed to assess the cost-effectiveness of the interventions. A total of 5 RCTs (with 904 adult participants) and 8 non-randomized studies (with 4,915 adult participants) assessing the use of the Body Composition Monitor [(BCM) Fresenius Medical Care, Bad Homburg vor der Hohe, Germany] were included. Both absolute over-hydration and relative over-hydration were significantly lower in patients evaluated using BCM measurements than for those evaluated using standard clinical methods [weighted MD [WMD] -0.44, 95 % CI: -0.72 to -0.15, $p = 0.003$, $I^2 = 49$ %; and WMD -1.84, 95 % CI: -3.65 to -0.03; $p = 0.05$, $I^2 = 52$ %, respectively]. Pooled effects of bio-impedance monitoring on systolic BP (SBP) (MD -2.46 mmHg, 95 % CI: -5.07 to 0.15 mmHg; $p = 0.06$, $I^2 = 0$ %), arterial stiffness (MD -1.18, 95 % CI: -3.14 to 0.78; $p = 0.24$, $I^2 = 92$ %) and mortality (HR = 0.689, 95 % CI: 0.23 to 2.08; $p = 0.51$) were not statistically significant. The economic evaluation showed that, when dialysis costs were included in the model, the probability of bio-impedance monitoring being cost-effective ranged from 13 % to 26 % at a willingness-to-pay threshold of £20,000 per quality-adjusted life-year gained. With dialysis costs excluded, the corresponding probabilities of cost-effectiveness ranged from 61 % to 67 %. The authors concluded that BCM used in addition to clinical assessment may lower over-hydration and potentially improve intermediate outcomes, such as SBP, but effects on mortality have not been demonstrated. They stated that if dialysis costs are not considered, the incremental cost-effectiveness ratio fell below £20,000, with modest effects on mortality and/or hospitalization rates, and the current findings are not generalizable to pediatric populations nor across other multi-frequency bio-impedance devices. These researchers stated that services that routinely use the BCM should report clinically relevant intermediate and long-term outcomes before and after introduction of the device to extend the current evidence base. The main drawbacks of this review were the lack of evidence on clinically relevant outcomes, children receiving dialysis, and any multi-frequency bio-impedance devices, other than the BCM.

Guidance from the National Institute for Health and Care Excellence (NICE, 2017) concluded: "[t]here is currently not enough evidence to recommend the routine adoption of the BCM – Body Composition Monitor to guide fluid management in people with chronic kidney disease having dialysis in the NHS. Further research is recommended to show the effect of using the BCM – Body Composition Monitor on clinical outcomes. The guidance also concluded: "[t]here is currently not enough validation or clinical-outcome data to recommend the routine adoption of the InBody S10 or the MultiScan 5000 to guide fluid management in people with chronic kidney disease having dialysis in the NHS."

Nasal Antibiotic for the Prevention of Peritonitis in Peritoneal Dialysis Individuals

Campbell and colleagues (2017) noted that PD is an important therapy for patients with

ESKD and is used in more than 200,000 such patients globally. However, its value is often limited by the development of infections such as peritonitis and exit-site and tunnel infections. Multiple strategies have been developed to reduce the risk of peritonitis including antibiotics, topical disinfectants to the exit site and anti-fungal agents. However, the effectiveness of these strategies has been variable and are based on a small number of RCTs. The optimal preventive strategies to reduce the occurrence of peritonitis remain unclear. This is an update of a Cochrane review first published in 2004. These investigators evaluated the benefits and harms of anti-microbial strategies used to prevent peritonitis in PD patients. They searched the Cochrane Kidney and Transplant's Specialized Register to October 4, 2016 through contact with the Information Specialist using search terms relevant to this review. Studies contained in the Specialized Register are identified through search strategies specifically designed for CENTRAL, Medline, and Embase; hand-searching conference proceedings; and searching the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov. Selection criteria were RCTs or quasi-RCTs in patients receiving chronic PD, which evaluated any anti-microbial agents used systemically or locally to prevent peritonitis or exit-site/tunnel infection. Two authors independently assessed risk of bias and extracted data. Summary estimates of effect were obtained using a random-effects model, and results were expressed as risk ratio (RR) with 95 % CI. A total of 39 studies, randomizing 4,435 patients, were included; 20 additional studies have been included in this update. The risk of bias domains were often unclear or high; risk of bias was judged to be low in 19 (49 %) studies for random sequence generation, 12 (31 %) studies for allocation concealment, 22 (56 %) studies for incomplete outcome reporting, and in 12 (31 %) studies for selective outcome reporting. Blinding of participants and personnel was considered to be at low risk of bias in 8 (21 %) and 10 studies (26 %) for blinding of outcome assessors. It should be noted that blinding of participants and personnel was not possible in many of the studies because of the nature of the intervention or control treatment. The use of oral or topical antibiotic compared with placebo/no treatment, had uncertain effects on the risk of exit-site/tunnel infection (3 studies, 191 patients, low quality evidence: RR 0.45, 95 % CI: 0.19 to 1.04) and the risk of peritonitis (5 studies, 395 patients, low quality evidence: RR 0.82, 95 % CI: 0.57 to 1.19). The use of nasal antibiotic compared with placebo/no treatment had uncertain effects on the risk of exit-site/tunnel infection (3 studies, 338 patients, low quality evidence: RR 1.34, 95 % CI: 0.62 to 2.87) and the risk of peritonitis (3 studies, 338 patients, low quality evidence: RR 0.94, 95 % CI: 0.67 to 1.31). Pre-/peri-operative intravenous vancomycin compared with no treatment may reduce the risk of early peritonitis (1 study, 177 patients, low quality evidence: RR 0.08, 95 % CI: 0.01 to 0.61) but has an uncertain effect on the risk of exit-site/tunnel infection (1 study, 177 patients, low quality evidence: RR 0.36, 95 % CI: 0.10 to 1.32). The use of topical disinfectant compared with standard care or other active treatment (antibiotic or other disinfectant) had uncertain effects on the risk of exit-site/tunnel infection (8 studies, 973 patients, low quality evidence, RR 1.00, 95 % CI: 0.75 to 1.33) and the risk of peritonitis (6 studies, 853 patients, low quality evidence: RR 0.83, 95 % CI: 0.65 to 1.06). Anti-fungal prophylaxis with oral nystatin/fluconazole compared with placebo/no treatment may reduce the risk of fungal peritonitis occurring after a patient has had an antibiotic course (2 studies, 817 patients, low quality evidence: RR 0.28, 95 % CI: 0.12 to 0.63). No intervention reduced the risk of catheter removal or replacement. Most of the available studies were small and of suboptimal quality; only 6 studies enrolled 200 or more patients. The authors concluded that in this update, they identified limited data from RCTs and quasi-RCTs that evaluated strategies to prevent peritonitis and exit-site/tunnel infections. This review demonstrated that pre-/peri-operative intravenous vancomycin may reduce the risk of early peritonitis and that anti-fungal prophylaxis with oral nystatin or fluconazole reduced the risk of fungal peritonitis following an antibiotic course. However, no other anti-microbial interventions have proven efficacy. In particular, the use of nasal antibiotic to eradicate *Staphylococcus aureus*, had an uncertain effect on the risk of peritonitis and raised questions about the usefulness of this approach. They stated that given the large number of patients on PD and the importance of peritonitis, the lack of adequately powered and high quality RCTs to inform decision-making about strategies to prevent peritonitis is striking.

Nicotinic Acid and Related Compounds for the Treatment of Hyperphosphatemia in Dialysis Persons

Liu and colleagues (2018) noted that studies indicated that nicotinic acid and related compounds may decrease phosphorus concentrations effectively by reducing the absorption in the gastro-intestinal (GI) tract. However, the efficacy and safety of oral niacin

treatments have only been investigated in a limited number of small-scale studies. These investigators performed a meta-analysis by pooling 12 qualified relevant pre-clinical and clinical trials to evaluate the association of nicotinic acid (and its related compounds) treatment and hyperphosphatemia among dialysis patients. Baseline and after treatment data were collected from the studies to evaluate drug efficacy, effect on lipid profile, and drug safety. To evaluate drug efficacy, subgroups were created based on different exposure time (i.e., 4 weeks, 8 weeks, 12 weeks, and 24 weeks) and each subgroup was compared against baseline data. In the assessment of lipid profile and drug safety, results of 8-week treatment were compared against baseline data. This study showed that in the efficacy assessment of drug treatment, serum phosphorus concentration was only significantly reduced in the 4-week (standardized mean difference [SMD], 0.68; 95 % CI: 0.40 to 0.97; $p = 0.000$; $n = 8$), and 8-week (SMD, 1.05; 95 % CI: 0.68 to 1.42; $p = 0.000$; $n = 10$) treatment groups. The calcium $\times$ phosphorus product showed significantly reduced concentration in all the drug exposure time settings, and no rebound was detected (4-week treatment: SMD, 0.61; 95 % CI: 0.18 to 1.04; $p = 0.005$; $n = 5$; 8-week treatment: SMD, 0.76; 95 % CI: 0.32 to 1.18, $p = 0.001$; $n = 8$; and 12-week treatment: SMD, 0.28, 95 % CI: -0.06 to 0.61; $p = 0.103$; $n = 3$). Lipid profile monitoring showed that HDL and triglycerides (TG) significantly changed after 8 weeks of treatment (HDL: SMD, -0.63; 95 % CI: -1.03 to 0.24; $p = 0.002$; $n = 5$) and TG: SMD, 0.25; 95 % CI: 0.02 to 0.49; $p = 0.033$; $n = 5$). Assessment of drug safety detected significant association for incidence of diarrhea (8 % incidence rate; 95 % CI: 4 % to 12 %; $p = 0.001$) and total AE (41 % incidence rate, 95 % CI: 12 % to 69 %, $p = 0.001$). The authors concluded that nicotinic acid and related compounds could significantly reduce serum phosphorus concentration with additive anti-lipemic effects. Moreover, they also recommended that the safety of this drug be further studied since these findings suggested significant incidence of AEs.

Peritoneal Dialysis for Heart Failure

Chionh and colleagues (2020) stated that heart failure (HF) is a major cause of morbidity and mortality. Extracorporeal (EC) therapy, including ultra-filtration (UF) and HD, peritoneal dialysis (PD) and peritoneal UF (PUF) are potential therapeutic options in diuretic-resistant states. In a systematic review, these researchers examined outcomes of PD and compared the effects of PD to EC. A comprehensive search of major databases from 1966 to 2017 for studies utilizing PD (or PUF) in diuretic-resistant HF was conducted, excluding studies involving patients with ESRD. Data were extracted and combined using a random-effects model, expressed as OR. A total of 31 studies ($n = 902$) were identified from 3,195 citations, none was randomized trials. Survival was variable (0 to 100 %) with a wide follow-up duration (36 hours to 10 years). With follow-up of greater than 1 year, the overall mortality was 48.3 %. Only 4 studies compared PD with EC. Survival was 42.1 % with PD and 45.0 % with EC; the pooled effect did not favor either (OR 0.80; 95 % CI: 0.24 to 2.69; $p = 0.710$). Studies on PD in patients with HF reported several benefits. Left ventricular ejection fraction (LVEF) improved after PD (OR 3.76, 95 % CI: 2.24 to 5.27; $p < 0.001$); 7 of 9 studies saw LVEF increase by more than 10 %; 21 studies reported the New York Heart Association (NYHA) status and 40 to 100 % of the patients improved by greater than or equal to 1 grade; 9 of 10 studies reported reductions in hospitalization frequency and/or duration. When treated with PD, HF patients had fewer symptoms, lower hospital admissions and duration compared to diuretic therapy. However, there was inadequate evidence comparing PD versus UF or HD. These researchers stated that further studies comparing these modalities in diuretic-resistant HF should be conducted.

Furthermore, UpToDate reviews on “Overview of the management of heart failure with reduced ejection fraction in adults” (Colucci, 2020), “Management of refractory heart failure with reduced ejection fraction” (Dunlay and Colucci, 2020), “Right heart failure: Causes and management” (Borlaug, 2020), and “Treatment and prognosis of heart failure with preserved ejection fraction” (Borlaug and Colucci, 2020) do not mention peritoneal dialysis as a management / therapeutic option.

American College of Cardiology guidelines (Hollenberg, et al., 2019) have no recommendation for peritoneal dialysis as a treatment for heart failure. The guidelines state that, for patients with volume overload refractory to diuretics, extracorporeal ultrafiltration or hemodialysis can be considered. The guidelines note that, although ultrafiltration and hemodialysis remove fluid effectively and can improve serum sodium, trials did not show improved clinical outcomes or kidney function.

Wearable Hemodialysis Devices

There is a lack of evidence supporting wearable hemodialysis devices. In a pilot study, Davenport and colleagues (2007) evaluated the safety and effectiveness of a wearable hemodialysis device. A total of 8 patients with ESRD (3 women and 5 men, mean age of 51.7 years) who were established on regular hemodialysis were fitted with a wearable hemodialysis device for 4 to 8 hours. Patients were given unfractionated heparin for anticoagulation, as they would be for standard hemodialysis. There were no important cardiovascular changes and no adverse changes in serum electrolytes or acid-base balance. There was no evidence of clinically significant hemolysis in any patient. Mean blood flow was 58.6 (SD 11.7) mL/min, with a dialysate flow of 47.1 (7.8) mL/min. The mean plasma urea clearance rate was 22.7 (5.2) mL/min and the mean plasma creatinine clearance rate was 20.7 (4.8) mL/min. Clotting of the vascular access occurred in 2 patients when the dose of heparin was decreased and the partial thromboplastin time returned towards the normal reference range in both of these patients. The fistula needle became dislodged in 1 patient, however safety mechanisms prevented blood loss, the needle was replaced, and treatment continued. The authors concluded that this wearable hemodialysis device shows promising safety and effectiveness results, although more research is needed to confirm these results.

A review of the evidence on the wearable hemodialysis device by the Australia and New Zealand Horizon Scanning Network (Mundy and Hiller, 2009) concluded: "Preliminary evidence of the wearable artificial kidney indicates that it is successful in the clearance harmful solutes and molecules that accumulate in patients with chronic kidney disease. The benefits of more frequent dialysis has been established and therefore a device which would enable patients to undergo dialysis frequently whilst able to participate in normal activities (including work) without being tied to a hospital setting would be advantageous both to the patient and to the health system. Studies where the WAK device is used long-term on a greater number of patients are required".

Vitamin E-Coated Membranes for Hemodialysis

Huang and associates (2015) noted that there is controversy regarding whether vitamin E-coated dialyzer therapy was beneficial for the complications associated with HD. These researchers performed a systematic review to evaluate the effects of vitamin E-coated dialyzer. Related trials were searched from multiple electronic databases. These investigators conducted a meta-analysis to evaluate changes in the pre-defined outcomes using RevMan 5.3 software. Meta-analysis showed vitamin E-coated dialyzer therapy could decrease erythropoietin (EPO) resistance index (standardized mean difference [SMD], -0.24; 95 % CI: -0.47 to -0.01; $p = 0.04$). However, pooled-analysis showed vitamin E-coated dialyzer therapy could not decrease weekly EPO dose (SMD, -0.11; 95 % CI: -0.32 to 0.09; $p = 0.28$) and intima-media thickness (IMT) of the carotid artery (MD, -0.09; 95 % CI: -0.2 to 0.01; $p = 0.09$), and vitamin E-coated dialyzer therapy did not improve the serum hemoglobin (MD, -0.03; 95 % CI: -0.18 to 0.13; $p = 0.74$), albumin levels (SMD, -0.64; 95 % CI: -1.62 to 0.34; $p = 0.2$), in addition, there was no significant difference in serum cholesterol (SMD, -0.07; 95 % CI: -0.45 to 0.31; $p = 0.71$), triglycerides (MD, -2.77; 95 % CI: -32.42 to 26.87; $p = 0.85$), high density lipoprotein (HDL) (SMD, 0.24; 95 % CI: -0.14 to 0.62; $p = 0.22$) and low density lipoprotein (LDL) (SMD, 0.00; 95 % CI: -0.38 to 0.37; $p = 0.98$) levels. The authors concluded that vitamin E-coated dialyzer may reduce the EPO resistance, but there was no conclusive evidence that vitamin E-coated dialyzer can improve the renal anemia, malnutrition, dyslipidemia and atherosclerosis status in HD patients. However, high-quality trials with hard clinical end-points are needed to fully elucidate the clinical value of vitamin E-coated dialyzer therapy.

D'Arrigo and colleagues (2017) stated that accruing evidence suggests that vitamin E-coated membranes (ViE-m) might improve the clinical management of chronic HD patients. These investigators conducted a systematic review and meta-analysis of RCTs comparing ViE-m to conventional HD; end-points were a series of biomarkers pertaining to anemia status, inflammation, oxidative stress and dialysis efficacy/status. A total of 60 studies were included; ViE-m significantly improved the Erythropoietin Resistance Index but had no impact on other anemia parameters. As for oxidative stress and inflammation, ViE-m produced a significant decrease in interleukin (IL)-6 levels, thiobarbituric acid reactive substances, plasma and red blood cell (RBC) malonylaldehyde and a significant increase in blood and RBC vitamin E. Conversely, ViE-m use had no impact on lipid profile, dialysis adequacy, blood pressure, albumin and uric acid. The authors concluded that ViE-m might ameliorated anemia management by reducing oxidative stress and inflammation. Moreover, they stated that benefits of these bio-membranes on harder clinical outcomes are uncertain and need to be investigated by future, targeted trials.

Hypoxia-Inducible Factor-Prolyl Hydroxylase Inhibitors Versus Erythropoiesis-Stimulating Agents on Iron Metabolism in Non-Dialysis-Dependent Anemic Patients with Chronic Kidney Disease

Sugahara et al (2022) noted that for the last 30 years, erythropoiesis-stimulating agents (ESA) in conjunction with iron supplementation has been the mainstay of treatment for anemia in CKD. Although ESAs are well-established and highly effective treatment, clinical trials showed that the use of ESAs with a high hemoglobin (Hb) target was associated with increased risk of cardiovascular events. This safety concern raised considerable interest in developing an alternative therapeutic strategy. Hypoxia-inducible factor-prolyl hydroxylase inhibitors (HIF-PHIs) are such novel agents employed for the treatment of anemia in CKD. They stimulate endogenous erythropoietin production via HIF activation; thus, inducing erythropoiesis. At least 6 small-molecule HIF-PHIs have been developed to-date. The phase-III clinical trials reported that their effects were non-inferior to ESAs. HIF-PHIs may have several advantages over the conventional treatment, such as oral route of administration and their ability to raise Hb levels in patients with chronic inflammation. Although many of the phase-III clinical trials showed that HIF-PHIs were non-inferior to placebo or ESAs with respect to cardiovascular safety, 1 of the compounds failed to meet the pre-specified non-inferiority criterion in non-dialysis-dependent (NDD)-CKD patients, and some studies of another HIF-PHI indicated potential risks for thromboembolic events. While the regulatory agencies of some countries including Japan and the European Union concluded that roxadustat, one of the HIF-PHIs, had a favorable benefit-risk profile, the Food and Drug Administration (FDA) decided not to approve the drug because of safety reasons. The authors concluded that to establish the optimal anemia management in CKD, further studies are needed to examine important aspects of HIF-PHIs, such as long-term safety, appropriate Hb target, and the types of patients who would gain benefits from these new drugs.

Wang et al (2022) stated that roxadustat is a newly marketed HIF-PHIs used to treat anemia in patients with ESRD. While clinical trials have showed the therapeutic effects of roxadustat in patients with ESRD who are resistant to ESAs, its metabolic effects are still unclear. A total of 32 individuals with ESRD and ESA resistance from the Blood Purification Center of Dalian Municipal Central Hospital were included. A total of 96 fasting serum samples were obtained from subjects before treatment with roxadustat, and after treatment for 15 and 30 days. Ultra-high performance liquid chromatography-mass spectrometry (HPLC-MS)-based metabolomics and lipidomics strategies were employed to examine the effects of roxadustat on serum metabolism. A total of 255 metabolites and 444 lipid molecular species were detected and quantified. Sphingolipids and phospholipids decreased significantly during treatment, possibly associated with changes in phospholipid and ceramide metabolism. Bile acid levels decreased and cholic acid/chenodeoxycholic acid increased, indicating changes in gut microbiota and bile acid metabolism. Amino acids also changed during the process of treatment. The authors concluded that the findings of this study showed sphingolipids, phospholipids, and bile acids were significantly altered, which may be associated with a changed metabolism caused by roxadustat. This approach provided a powerful tool for examining the mechanisms of ESA resistance in ESRD patients and may represent a promising strategy for elucidating the complex therapeutic mechanisms of other drugs. These researchers stated that further studies are needed to confirm these findings.

The authors stated that this study had several drawbacks. These researchers only examined 96 samples from 32 subjects and did not validate the metabolomics and lipidomics results in additional samples. In addition, the proposed mechanism was not verified by other experiments, such as proteomics, cell lines, or animal models.

In a network meta-analysis, Yang et al (2023) compared the effects of 5 HIF-PHIs, 2 ESAs, and placebo on iron metabolism in renal anemia patients with NDD-CKD. A total of 5 electronic databases were searched for studies; RCTs comparing HIF-PHIs, ESAs, and placebo in NDD-CKD patients were selected. The statistical program used for network meta-analysis was Stata/SE 15.1. The main outcomes were the change in hepcidin and Hb levels. The merits of intervention measures were predicted by the surface under the cumulative ranking curve method. Of 1,589 original titles screened, data were extracted from 15 studies (3,228 participants). All HIF-PHIs and ESAs showed greater Hb level-raising ability than placebo. Among them, desidustat showed the highest probability of increasing Hb (95.6 %). Hepcidin [mean deviation = -43.42, 95 % CI: -47.08 to -39.76], ferritin (MD = -48.56, 95 % CI: -55.21 to -41.96), and transferrin saturation (MD = -4.73, 95 % CI: -5.52 to -3.94) were decreased, while transferrin (MD = 0.09, 95 % CI:

0.01 to 0.18) and total iron-binding capacity (MD = 6.34, 95 % CI: 5.71 to 6.96) was increased in HIF-PHIs versus those in ESAs. Furthermore, this study observed heterogeneity in the ability of HIF-PHIs to decrease hepcidin. Compared with darbepoetin, only daprodustat (MD = -49.09, 95 % CI: -98.13 to -0.05) could significantly reduce hepcidin levels. Meanwhile, daprodustat also showed the highest hepcidin-lowering efficacy (84.0 %), while placebo was the lowest (8.2 %). The authors concluded that the findings of this study suggested that HIF-PHIs should be the 1st choice in the treatment of NDD-CKD patients with functional iron deficiency. In addition, the potential heterogeneity of HIF-PHIs to correct iron metabolism should be fully considered when formulating a treatment plan.

The authors stated that this network meta-analysis had several drawbacks. First, some of the included studies did not use the double-blind method, but, considering that the included outcomes of this meta-analysis had a unified standard and were not easy to be changed by subjective factors, non-double-blind trials were not excluded. Second, the follow-up time between the included studies was quite different; therefore, this meta-analysis could not accurately observe the iron metabolism of patients in different administration stages. However, considering that most HIF-PHIs have not yet been approved for clinical use, this analysis still included some phase-II clinical trials with shorter dosing time. Third, there were differences in the CKD status of the subjects in the included trials, which might make their iron metabolism status different. However, after calculation, it was found that this difference did not meet the statistical criteria for heterogeneity; therefore, this study did not adjust the inclusion and exclusion criteria.

Chen et al (2023) stated that 5 types of HIF-PHIs have been authorized for anemia treatment in CKD patients in China and Japan -- enarodustat, roxadustat, daprodustat, vadadustat, and molidustat. How effectively they compared to ESAs about clinical results in CKD-DD patients is uncertain. In a network meta-analysis, these investigators examined the RCT evidence regarding the risks and benefits of HIF-PHIs and ESAs in CKD patients on dialysis. In these RCTs, patients with CKD-DD received 1 of 5 different HIF-PHI or ESAs, a placebo, and no medical intervention. Outcomes included Hb, iron parameters, and AEs, and there were 4 weeks of follow-up at least. A frequentist framework for multi-variate random effects meta-analyzed the results. The effect sizes of categorical variables were displayed as ORs; MDs were used for calculating continuous outcomes with common units; otherwise, SMDs were applied. The Cochrane tool evaluated the bias risk in RCTs. A total of 26 RCTs with 1,4945 patients were qualified for inclusion. Compared to the placebo, HIF-PHIs and ESAs dramatically boosted Hb without affecting serum iron. Roxadustat resulted in better Hb levels than ESAs (MD 0.32, 95 % CI: 0.10 to 0.53), and daprodustat (0.46, 0.09 to 0.84). Roxadustat (91.8 %) was the top Hb treatment among all medical interventions, as determined by the SUCRA ranking. However, roxadustat caused more thrombosis and hypertension than ESAs (1.61, 1.22 to 2.12) and vadadustat (1.36, 1.01 to 1.82). The lowest rates of hypertension and thrombosis were observed in molidustat (80.7 %) and ESAs (88.5 %). Compared with a placebo, ESAs and HIF-PHIs all affected TSAT levels. Except for molidustat, the other 4 HIF-PHIs impacted different iron parameters. Regarding ferritin reduction, roxadustat (90.9 %) and daprodustat (60.9 %) were the top 2 agents. Enarodustat (80.9 %) and roxadustat (74 %) placed 1st and 2nd in lowering hepcidin levels. The former 2 medicines for TIBC improvement were vadadustat (98.7 %) and enarodustat (80.9 %). The authors concluded that the most effective treatment for correction of Hb was roxadustat. The superior effectiveness of reducing hepcidin made roxadustat and enarodustat appropriate for patients with inflammation. However, the increased risk of hypertension and thrombosis associated with roxadustat should be noted. In patients at risk for hypertension and thrombosis, molidustat and ESAs may be preferable options. When administering roxadustat and daprodustat, clinicians should check ferritin to examine iron storage. Moreover, these researchers stated that lower TSAT in patients receiving HIF-PHIs and ESAs treatment suggested intravenous (IV) iron supplements are needed.

The authors stated that this study had 2 main drawbacks. First, the findings of this study came from CKD-DD participants. It is not suitable for CKD-NDD patients' applications. Second, whether iron supplements should be prescribed in CKD-DD patients during HIF-PHIs intervention is unknown.

Interleukin-6 Levels for Estimation of Cardiovascular and All-Cause Mortality Risk in Dialysis Patients

Chen and Wang (2023) stated that although previous studies have examined the correlation

of interleukin-6 (IL-6) with mortality risk in dialysis patients, the findings have been conflicting. In a meta-analysis, these investigators examined the use of IL-6 measurement for estimating cardiovascular mortality and all-cause mortality in dialysis patients. Embase, PubMed, Web of Science, and Medline databases were searched to identify relevant studies. After screening out the eligible studies, the data were extracted. A total of 28 eligible studies with 8,370 dialysis patients were included. Pooled analyses revealed that higher IL-6 levels were related to increased cardiovascular mortality risk (HR = 1.55, 95 % CI: 1.20 to 1.90) and all-cause mortality risk (HR = 1.11, 95 % CI: 1.05 to 1.17) in dialysis patients. Further subgroup analyses suggested that higher IL-6 levels were associated with elevated cardiovascular mortality in hemodialysis patients (HR = 1.59, 95 % CI: 1.36 to 1.81) but not in peritoneal dialysis patients (HR = 1.56, 95 % CI: 0.46 to 2.67). Moreover, sensitivity analyses indicated that the results were robust. Egger's test revealed potential publication bias among studies examining the correlation of IL-6 levels with cardiovascular mortality ($p = 0.004$) and all-cause mortality ($p < 0.001$); however, publication bias was not observed when using Begg's test (both $p > 0.05$). The authors concluded that this meta-analysis showed that higher IL-6 levels could indicate higher risks of cardiovascular mortality and all-cause mortality in dialysis patients. These findings suggested that monitoring IL-6 cytokine may help to enhance dialysis management and improve the general prognosis of patients.

These researchers stated that publication bias affected this study and served as one of the major drawbacks. Other inherent drawbacks of meta-analyses also affected the present trial, such as the existence of heterogeneity; thus, limiting the generalizability of these findings. Furthermore, most included studies in the current meta-analysis were observational in nature; therefore, some confounding factors might still exist; and the actual risk could be under-estimated.

Prophylactic Peritoneal Dialysis for Improvement of Post-Operative Outcomes in Infants/Children Undergoing Cardiac Surgery

Namachivayam et al (2022) stated that peritoneal dialysis (PD) is used in several cardiac surgical units following cardiac surgery, and early initiation of PD after surgery may have the potential to influence post-operative outcomes. In a systematic review and meta-analysis, these investigators examined the available evidence for the association between early PD after cardiac surgery and post-operative outcomes. Data sources included Medline, Embase, and PubMed from 1981 to November 1, 2021. Observational studies and randomized trials reporting on early PD after pediatric cardiac surgery were selected for analysis. Random-effects meta-analysis was used to estimate the pooled ORs and their 95 % CIs for post-operative mortality and pooled mean difference (MD) (95 % CI) for duration of mechanical ventilation and intensive-care unit (ICU) length of stay (LOS). These investigators identified 9 studies from the systematic review, and 5 were considered suitable for meta-analysis. Early initiation of PD after cardiac surgery was associated with a reduction in post-operative mortality (OR, 0.43; 95 % CI: 0.23 to 0.80; number of estimates = 4). Early commencement of PD shortened duration of mechanical ventilation (MD [95 % CI], -1.09 days [-1.86 to -0.33 d]; $I^2 = 56.1$ %; $p = 0.06$) and ICU-LOS (MD [95 % CI], -2.46 days [-3.57 to -1.35 d]; $I^2 = 18.7$ %; $p = 0.30$), respectively. All 3 estimates had broad 95 % prediction intervals (crossing null) denoting major heterogeneity between studies and wide range of possible study estimates in similar future studies. Overall, studies reporting on the effects of early PD included only a subset of infants undergoing cardiac surgery (typically high-risk infants), so selection bias may be a major issue in published studies. The authors concluded that the findings of this review suggested that early initiation of PD may be associated with beneficial post-operative outcomes in infants following cardiac surgery; however, these findings were based on studies of varying qualities and risk of bias. Early identification of high-risk infants following cardiac surgery was important so that prevention or early mitigation strategies could be applied to this cohort. Moreover, these researchers stated that future prospective studies in high-risk populations are needed to examine the role of early PD in influencing post-operative outcomes.

In a systematic review and meta-analysis, Ulrich et al (2024) examined if prophylactic PD catheter insertion at the time of pediatric cardiac surgery is associated with improved short-term outcomes. Databases search of the Medline, Embase, CINAHL, and Cochrane Library were completed in April 2021 and updated in October 2023. Two reviewers independently completed study selection, data extraction, and bias assessment. Inclusion criteria were RCTs and observational studies of children (18 years of age or younger)

undergoing cardiac surgery with cardiopulmonary bypass. They examined the use of prophylactic PD catheter versus no prophylaxis. The primary outcome was in-hospital mortality, as well as secondary short-term outcomes. Pooled random-effect meta-analysis OR with 95 % CI were reported. A total of 17 studies met inclusion criteria, entailing 4 RCTs. The non-PD catheter group received supportive care that included diuretics and late placement of PD catheters in the ICU. Most study populations included children younger than 1 year and weight less than 10 kg. Cardiac surgery was most commonly used for arterial switch operation. In-hospital mortality was reported in 13 studies; pooled analysis showed no association between prophylactic PD catheter placement and in-hospital mortality. There were mixed results for ICU-LOS and time to negative fluid balance, with some studies showing shortened duration associated with use of prophylactic PD catheter insertion and others showing no difference. Overall, the studies had high risk for bias, mainly due to small sample size and lack of generalizability. The authors concluded that in this meta-analysis failed to show an association between prophylactic PD catheter insertion in children and infants undergoing cardiac surgery and reduced in-hospital mortality. Other relevant short-term outcomes, including markers of fluid over-load, require further investigations.

The EchoSure System

Calotta et al (2023) stated that arterio-venous fistulae (AVF) are considered the preferred hemodialysis access; however, up to 50 % of all AVF created in the U.S. never mature. Doppler ultrasound (DUS) is useful for predicting fistula maturity and impending fistula failure. DUS is resource-intensive and is associated with poor compliance rates in dialysis patients, ranging from 12 % to 33 %. EchoSure is an FDA-cleared 3D Doppler US device that automatically delivers quantitative blood flow and anatomic vascular information. The technology can be used at the bedside by personnel without formal sonographic training, nullifying limitations of traditional Duplex US imaging. These researchers compared the EchoSure system in the hands of inexperienced personnel to a traditional expert-operated DUS for rapid evaluation of a bench-top model vascular system with flow, diameter, and depth expected in a human AVF. Both Duplex and EchoSure carried out within the expected tolerance of US readings (35 %) for volume flow, with the average error (AE) between the observed measurement and the ground truth being 8 % for Duplex, and 8 % for EchoSure. However, the average coefficient of variation (CV) for Duplex pooled over all flow rate measurements was 17 % versus 4 % for EchoSure. Regarding diameter, Duplex measurements had AE of 15 % with an average CV of 6 % across all measurements versus EchoSure AE of 4 % and average CV of 2 %. Duplex and EchoSure measurements over all depths had the same AE of 2 %. The 2 modalities were not statistically different for depth measurement ($p = 0.05$); however, EchoSure measured closer to the ground truth for flow rate and vessel diameter (flow: $p = 0.028$, $p = -0.07$; diameter: $p < 0.001$, $p = 0.69$). The inexperienced personnel using EchoSure acquired data 62 % faster than the expert sonographers using the Duplex US (141 mins for Duplex versus 87 mins for EchoSure). The authors concluded that the EchoSure platform may offer a safe, accurate, and convenient alternative to Duplex ultrasonography for imaging fistulas in the dialysis clinic. Moreover, these researchers stated that clinical studies to define the role of EchoSure in the care of end stage renal disease (ESRD) patients with AVF are needed.

The authors stated that this study had 2 main drawbacks. First, this was a bench-top model of AVFs rather than a clinical study; therefore, additional clinical validation is needed. Second, the benchtop model used in the experiment is novel and has not been validated as correlating to clinical AVF outcomes. These investigators noted that despite these drawbacks, the EchoSure System appeared to have potential as a tool to acquire critical AVF decision-making data in the dialysis unit, and offers several advantages over DUS. For patients, the examination time is much shorter and does not require extra appointments and the associated travel and financial burden.

Percutaneous Hemodialysis Arterio-Venous (AV) Fistulas

Wasse et al (2020) noted that ESKD patients who are candidates for surgical AV fistula creation commonly experience obstacles to a functional surgical AV fistula, including protracted wait-time for creation, poor maturation, and surgical AV fistula dysfunction that can result in significant patient morbidity. The recent approval of 2 endovascular devices designed to create a percutaneous AV fistula enables AV fistula creation to be placed in the hands of interventionalists; thus, increasing the number of AV fistula providers, reducing wait-times, and allowing the patient to avoid surgery. Moreover,

current studies show that patients with percutaneous AV fistula experience improved time to AV fistula maturation.

Rajan (2023) stated that non-surgical, percutaneous, or endovascular hemodialysis AV creation represent an evolution of access creation away from traditional surgical fistulas. These fistulas are additional to surgical alternatives and published studies with the 2 commercially available devices suggested positive outcomes in terms of technical success, maturation, functionality, and patency.

In a systematic review, Bontinis et al (2023) examined the short-term and mid-term safety and effectiveness of percutaneous endovascular AV fistula (pAVF) creation. This systematic search was implemented corresponding to the PRISMA 2020 and the PRISMA for individual participant data (IPD) systematic reviews 2015. Aggregated data from the included studies were obtained and meta-analyzed regarding both the overall pAVF effectiveness and the comparison of pAVF with surgical AVF (sAVF). These investigators carried out a 2-stage IPD meta-analysis for studies comparing pAVF and sAVF regarding primary and secondary patency. Primary end-points included primary patency, secondary patency, and functional cannulation. A total of 18 studies with 1,863 patients were included. The overall pAVF, primary patency, secondary patency, functional cannulation and abandonment rates were 54.01 % (95 % CI: 40.69 to 66.79), 87.27 % (95 % CI: 81.53 to 91.42), 79.94 % (95 % CI: 65.94 to 89.13), and 15.58 % (95 % CI: 7.77 to 28.79), respectively. The overall pAVF, technical success, maturation, re-intervention per person-years and mean time to maturation rates were 97.08 % (95 % CI: 95.66 to 98.04), 82.13 % (95 % CI: 71.64 to 89.32), 0.80 (95 % CI: 0.34 to 1.47), and 58 days (95 % CI: 36.64 to 92.82), respectively. Secondary patency and pAVF abandonment rates where the only end-points were WavelinQ and Ellipsys displayed statistically significant differences of 81.36 % (95 % CI: 76.15 to 85.65) versus 92.12 % (95 % CI: 87.94 to 94.93) and 32.54 % (95 % CI: 22.23 to 44.87) versus 11.13 % (95 % CI: 4.82 to 23.65). An IPD meta-analysis of hazard ratios (HRs) for primary and secondary patency between pAVF and sAVF were 1.27 (95 % CI: 0.61 to 2.67) and 1.25 (95 % CI: 0.87 to 1.80), favoring sAVF.

Statistically significant difference between pAVF and sAVF were solely depicted for steal syndrome RR of 5.91 (95 % CI: 1.12 to 31.12) and wound infections RR of 4.19 (95 % CI: 1.04 to 16.88). Plotting of pAVF smoothed hazard estimate displayed an up-surge in the probability of primary patency failure at 1 month after the intervention. The authors concluded that although they failed to identify statistically significant differences between pAVF and sAVF regarding any of the primary end-points, pAVF displayed a decreased risk for steal syndrome and wound infection. Although both the Ellipsys and WavelinQ devices displayed satisfactory secondary patency rates, Ellipsys showed a statistically significant improved rate compared with WavelinQ. Furthermore, and despite the borderline statistically insignificant inferior re-intervention rate displayed by WavelinQ, 1 in 3 WavelinQ pAVFs resulted in abandonment. The introduction of pAVF as a treatment modality calls for standardized definition adjustment and improvement.

Furthermore, an UpToDate review on "Percutaneous hemodialysis arteriovenous fistula" (Smeds and Bose, 2024) states that "Percutaneous hemodialysis arteriovenous (AV) fistulas, which are also termed endo-AV fistulas or endoAVFs, are created using endovascular techniques without the need for a surgical incision. In the absence of anatomy acceptable for distal surgical AV fistula (e.g., radial-cephalic fistula), an endoAVF is an option for AV fistula creation ... A variety of studies have evaluated outcomes percutaneous AV fistulas, with generally favorable short-term results.

Percutaneous AV fistulas have high initial technical success rates and reasonable maturation rates. No clinically important differences have been proven when comparing the available percutaneous AV fistula systems. Compared with surgical AV fistulas, patency rates appear similar out to two years".

VasQ Extravascular Support Device for Improvement of Surgical Fistula Outcomes

The VasQ nitrol extravascular support device (Laminate Medical Technologies Ltd, Tel Aviv, Israel) is designed for the improvement of surgical fistula outcomes by providing permanent reinforcement to the artery-vein connection, promoting stability against wall tension, and decreasing turbulent flow.

Leonardi et al (2021) stated that the external support device VasQ is intended to promote arterio-venous fistula (AVF) maturation by maintaining the optimal anastomotic angle in order to minimize blood flow disturbances around the anastomotic area. In a prospective, single-center study, these researchers examined the safety and effectiveness of the VasQ device both in brachio-cephalic (BC) and radio-cephalic (RC) fistulae. This trial was

carried out with implantation of the VasQ device during AVF creation. Clinical and Doppler ultrasound (US) evaluations were carried out at day-1, and at 1, 6, and 12 months for assessment of device-related AEs, AVF maturation and patency. Moreover, volume flow rate and diameter of outflow vein were measured. A total of 16 patients were enrolled -- 10 BC-AVFs, and 6 RC-AVFs were created with VasQ. Pre-operatively cephalic vein diameter was 3.6 ± 0.9 mm. This trial included 13 men and 3 women, 9 end stage kidney disease (ESKD) in conservative therapy, 4 dialysis-treated patients, and 3 transplanted patients; mean age was 74.0 ± 8.1 years; no severe device-related AEs were observed. Primary patency (PP) at 1, 6, and 12 months was 100 %, 87.5 %, and 67.7%, respectively, while secondary patency (SP) was 100 %, 100 %, and 78.3 %, respectively. Comparing BC-AVFs RC-AVFs, no significant differences in PP or SP rates were observed. Overall maturation rate was 94 % (15/16). Mean vein diameter measured with Doppler US at post-operative day-1 and at 1, 6, and 12 months was 5.0 ± 1.0 , 5.9 ± 0.9 , 7.2 ± 1.6 , and 7.9 ± 1.4 mm, respectively, with a mean flow rate at the brachial artery of 841 ± 176 , $1,052 \pm 224$, 1261 ± 490 , and $1,348 \pm 477$ ml/min, respectively. The authors concluded that in their limited experience VasQ was safe, with high maturation and patency rates; positive results suggested a potential benefit for VasQ in AVF.

In a retrospective, multi-national study, Shahverdyan et al (2022) examined the outcomes of AVF, created with VasQ external support device under standard clinical practice across 3 vascular access clinics. These researchers evaluated prospectively collected clinical outcomes of both fore-arm and upper-arm BC-AVF created using the VasQ device. A total of 51 AVF patients were pooled from 3 centers in Germany, Italy, and Spain. Post-procedure outcomes were evaluated by flow measurement with US examination in all centers. Primary failure, usability, patency, and intervention rates during the study period were evaluated. A total of 51 VasQ devices were implanted during 37 fore-arm (36 RC-AVFs and 1 ulnar-basilic AVF) and 14 BC-AVFs procedures. The study population comprised mostly of men (73 %) and Caucasians (82 %), with a mean age of 62.5 years (range of 38 to 84 years), and mean body mass index (BMI) of 29.2. One patient died due to access-unrelated reasons, and 1 patient was lost to follow-up without AVF assessment after its creation. From the remaining 49, 3 patients had a follow-up of less than 4 weeks and were excluded from maturation, primary failure, and dialysis initiation analysis (because those factors could not be evaluated yet), leaving 46 patients in this evaluation. Maturation was achieved in 91 % of patients. The primary failure rate was 9 %. A PP rate of 77 % was observed at 6, 12, and 18 months. The SP rate was 91 % at 6 months, and 85 % at 12 months and 18 months. Both PP and SP rates did not differ statistically when comparing fore-arm AVF with BC-AVF ($p > 0.25$). Successful cannulation was achieved in 86 % of patients (32/37) requiring dialysis with a median time from creation to cannulation of 46 days. Of those patients, the functional patency rate was 94 %; 5 patients underwent 7 interventions to maintain or restore patency or functionality, with an overall rate of 0.248 interventions per patient-year. The authors concluded that AVFs created with VasQ external support device demonstrated promising patency and functionality outcomes in multi-centric real-world setting across the 2 most commonly used anatomic locations.

The authors stated that this study had the drawbacks of any retrospective analysis. It relied on medical charts and medical history and as such could be affected by the quality of the reported data. The unbalanced recruitment of patients from different vascular centers might have affected the pooled data. To minimize this risk, these investigators have contacted the dialysis units treating the patients and verified the patient data. Another drawback might be the relatively high rate of patients who were pre-dialysis at the time of fistula creation.

In a single-center study, Benedetto et al (2022) compared the maturation rate of the native RC-AVF created with and without VasQ. Data of all consecutive patients who underwent the creation of native RC-AVFs at the authors' center between October 2018 and January 2020 was prospectively collected and retrospectively analyzed. Selected patients who had a suitable vein and a radial artery with tri-phasic flow at pre-operative duplex US examination and were selected for the creation of a RC-AVF were included. Exclusion criteria included malignant tumors, acute renal failure, previous upper limb re-vascularization, as well as septic status. Patency and maturation, vein, and artery diameter, and blood flow rate were evaluated at the following intervals: post-operatively, 24 hours post-operatively, 1 month, 3 months, and 6 months post-operatively. A total of 49 patients (31 men, mean age of 65.7 years) were included in this trial; 25 subjects received VasQ devices (VasQ group), the remaining 24 subjects formed the control group. All participants underwent RC-AVF placement (21 on the wrist, 20 on the fore-arm, 8 on the proximal fore-arm). There were no peri-operative complications and fatalities. At 1, 3, and 6 months, PP rates were $96 \% \pm 4 \%$, $96 \% \pm 4$

%, $91\% \pm 6\%$ (VasQ group) versus $87\% \pm 7\%$, $87\% \pm 7\%$, $80\% \pm 9\%$ (control group, $p = 0.17$); SP rates were $96\% \pm 4\%$, $96\% \pm 4\%$, $91\% \pm 6\%$ (VasQ group) versus $95\% \pm 4\%$, $90\% \pm 7\%$, $90\% \pm 7\%$ (control group, $p = 0.79$). A significantly larger vein diameter increase post-operatively ($p < 0.009$) and a greater maturation rate (96% versus 74% , $p < 0.044$) were found in the VasQ group compared to the control group. The authors concluded that the use of the VasQ device was associated with higher maturation rates and larger vein diameters post-operatively. The patency rates were slightly higher but not significantly. These researchers stated that further studies are needed to confirm these findings.

Dillavou et al (2023) noted that AVF creation is a frequently performed vascular procedure that reported 6-month functional success rates as low as 50% . Recently, VasQ, has shown potential in studies outside the U.S. to improve AVF outcomes when implanted at creation. These investigators discussed the pivotal study results of this novel technology in treating patients in the U.S. VasQ was implanted in 144 patients at 16 centers across the U.S. who were referred for creation of a new AVF and consented for enrollment in a 2-year, prospective, open-label, single-arm, multi-center study. BC-AVFs ($n = 129$) and RCV-AVFs ($n = 15$) were analyzed. The primary endpoint was PP at 6 months compared against a performance goal of 55% derived from a systematic literature search. Safety endpoints included device-related events, ischemic steal, infection, aneurysm, and seroma at up to 6 months. Minimum arterial size was 2.0 mm ; target veins were required to measure 2.5 to 6 mm . Key exclusion criteria were patients less than 18 years of age or greater than 80 years of age, those with known ipsilateral central venous occlusion, target cannulation zone venous depth greater than 8 mm , and NYHA functional class of 3 or 4. Patients were 61% men, 53% White, 35% African American, and 14% Hispanic. Mean age was 60 years, and median BMI was 30.4 . Of the patients, 69% were diabetic, 66% were on dialysis at the time of creation, and 70% had a prior access surgery. At 6 months, steal was observed in 2.1% , infection in 0.7% , and no aneurysms or seromas were observed. PP rate at 6 months was 66% ($p < 0.021$ versus performance goal). Physiological maturation was achieved in 92.4% of patients. Successful 2-needle cannulation for patients who entered the study on dialysis was achieved in 88% of VasQ AVFs at a median of 56 days. Pre-dialysis patients who initiated dialysis during the study achieved 2-needle cannulation in 81.6% VasQ AVFs. Interventions were required at a rate of 1.07 per patient year over the entire study period. Two-year cumulative patency was 76.6% (95% CI: 67.9% to 83.4%) with no statistical difference between patients requiring interventions and those that did not. No patency differences were observed between BC-AVFs and RC-AVFs. The authors concluded that the findings of the U.S. pivotal study showed improved AVF outcomes and an excellent safety profile with VasQ use relative to traditional AVFs. These researchers stated that under the conditions of this trial, VasQ demonstrated great promise in expeditiously and efficiently enhancing AVF functional success.

The authors stated that this trial had several drawbacks. First, the use of a systematic literature review to develop a performance objective for the primary endpoint. A direct control group would reduce the uncertainty of potential confounding factors that may have resulted in the improved outcomes observed for the VasQ patients when compared across reported outcomes for traditional AVFs. Second, statistical comparisons were only carried out for the primary endpoint, leaving all other comparisons to be qualitative in nature. Third, the study excluded patients that would require a transposition, leaving open the question if the benefits observed for VasQ AVFs would extend to that sub-population.

Bozzetto et al (2024) stated that AVF is the preferred vascular access (VA) for hemodialysis; however, it is still affected by high non-maturation and early failure rates due to stenosis development. Increasing evidence suggests that the presence of turbulent-like flow may play a key role. Thus, to stabilize the flow in the venous segment, an external support device (VasQ) has been designed. In a prospective, randomized, single-center, pilot study, these researchers provided preliminary evidence on the impact of VasQ on AVF hemodynamics as compared to AVFs created with conventional surgery. A total of 6 patients were enrolled: 3 in the VasQ group and 3 in the control group. Contrast-free magnetic resonance imaging (MRI) scans were acquired at 3 days, 3 months, and 1 year after AVF surgery and were used to generate three dimensional (3D) patient-specific models. Computational fluid dynamic (CFD) simulations were carried out using pimpleFoam, imposing patient-specific flow wave-forms derived from US examinations at the inlet of the proximal and distal artery, and a traction-free condition at the venous outflow. Morphologic and hemodynamic changes occurring over time were compared between VasQ and control AVFs. MRI protocol provided high-quality images suitable for

reliable segmentation and reconstruction of patient-specific 3D models of AVFs at all 3 time-points in 4 of the 6 patients. The VasQ device maintained the angle between the artery and the vein almost unchanged over time, with a more stable flow in the AVFs supported by the device. In contrast, 1 of the AVFs of the control group evolved to an extreme dilatation of the vein and highly disturbed flow, while the other developed a stenosis in the juxta-anastomotic region. The authors concluded that the findings of this pilot study showed the feasibility of characterizing the morphological and hemodynamic changes occurring over time in AVFs created using the VasQ device, and provided preliminary evidence of the potential hemodynamic benefits of its use. Moreover, these researchers stated that future clinical investigations with higher number of patients are needed to confirm these preliminary findings, fully examine the role of VasQ device in flow regularization and elucidate its potential in preventing stenosis development and AVF failure.

The authors stated that the principal drawback of this trial was the small sample size ($n = 6$) that allowed only preliminary observations on the effect of VasQ. However, these investigators studied only 2 patients per group to examine the feasibility of comparing VasQ and control AVFs in a prospective study and in the long-term. Moreover, the study included only side-to-end AVFs but this was due to the fact that VasQ device was designed to create only this type of anastomosis. Another drawback was that CFD simulations did not account for the effect of the vessel wall compliance. This issue could be addressed in future studies by conducting fluid-structure interaction (FSI) simulations. McGah et al (2014) determined that rigid-wall hemodynamic simulations could predict blood hemodynamics within the same order of accuracy of the FSI equivalent simulations, but how the VasQ device would affect stresses in the vascular wall will be an interesting future development.

In a single-center study, Fontseré et al (2024) described their experience of implanting the VasQ device in the maturation of native AVF (nAVF) in a subgroup of patients (2018 to 2019). Socio-demographic, US, as well as re-vascularization procedure data were analyzed. For statistical analysis, descriptive techniques were employed: Chi-2 test for proportions and t-test for continuous variables. Kaplan-Meier survival curves were used to calculate PP and SP rates, and Cox regression analysis for subgroup comparison, using the SPSS® v. 21 statistical package. Significance was considered if $p \leq 0.05$. For the study of clinical and US maturation, the standards of the current Spanish vascular access guidelines were used. A total of 21 patients with stage-5 chronic kidney disease on dialysis (CKD-5D) were included (6 women and 15 men), with a mean age of 69.8 ± 13.2 years (26 to 85 years), 9 with RC-AVF and 12 with humero-cephalic nAVF over which the VasQ device was implanted. Adequate clinical and US maturation rates were achieved in 95.2 % of patients. There were no surgical complications of note or significant differences in the location of the nAVF (distal versus proximal). Pre-operative mapping showed a mean feeding artery of 0.31 ± 0.16 cm and vein of 0.31 ± 0.15 cm. The PP rates obtained at 1, 3, 6 and 12 months were 95.2 %, 90.5 %, 71.4 %, and 52.4 %, respectively, whereas the SP rates at the same time-points were 95.2 %, 90.5 %, 85.7 %, and 83 %, respectively. A total of 20 balloon-catheter percutaneous transluminal angioplasties (PTA) were carried out during the follow-up period, although 14 of these procedures were performed on just 3 of the patients. These investigators found no significant differences in patency when comparing these results with a similar small, retrospective cohort with isometric exercises (30 patients). The 6-month PP and SP rates reported by other VasQ device studies were similar to those obtained by this group (PP = 79 % to 87.5%; and SP = 79 % to 100 %). These researchers stated that the principal drawbacks of these investigations were the inclusion of few patients (less than 35), as well as the lack of randomization with a control group based on conventional surgery, which made interpretation of these findings difficult. The authors concluded that, based on these preliminary findings, the VasQ device was useful and safe, providing adequate maturation and good patency rates in nAVF. Moreover, these researchers stated that they are designing a prospective, randomised trial in patients with radio-cephalic (distal) nAVF to examine if this new device would represent a cost-effective advance in the field of vascular access.

Li et al (2025) noted that VasQ is an external support device for autologous AVF designed to improve anastomotic blood flow and reduce neointimal hyperplasia; however, different studies have shown that the effectiveness of the VasQ device in improving AVF is inconsistent. In a meta-analysis, these investigators examined the safety and effectiveness of the VasQ device. Two reviewers independently searched studies published in PubMed, Embase, Cochrane, Web of Science, CNKI, and Wan Fang databases from inception to 2023. The Cochrane Systematic Evaluation Bias Risk Tool Version 1 was used to examine

the risk of RCTs bias. The ROBINS-I tool was used to examine the risk of bias in non-randomized studies. A single-arm meta-analysis was carried out, and a random effects model was used for all analyses. These researchers identified 6 studies entailing 146 patients and carried out a meta-analysis. The results showed that after 6 months of VasQ device treatment, the PP rate of AVF was (76.4 %, 95 % CI: 0.608 to 0.920, $p < 0.01$); while the SP rate was 76.5 % (95 % CI: 0.572 to 0.958, $p < 0.01$). The maturity rate of AVF 1 month after surgery was 88.5 % (95 % CI: 0.818 to 0.952, $p = 0.46$). The incidence of anastomotic stenosis was 8.9 % (95 % CI: 0.015 to 0.163, $p = 0.23$), and the incidence of anastomotic venous thrombosis was 10 % (95 % CI: 0.035 to 0.179, $p = 0.38$). The authors concluded that meta-analysis data of this study showed that the VasQ device exhibited a good effect in improving the patency rate of AVF and did not increase the occurrence of AEs; however, due to the limitation of the number and quality of included studies, more high-quality studies are needed to confirm these findings.

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